



Acoustic and thermophysical aspects of AeZrH₃ Perovskites for efficient hydrogen storage via d-band center: DFT+U study with van der Waals interactions

M. Kashif Masood^{a,*}, Rania Charif^{b,**}, Redi Kristian Pingak^c, Asif Jamil^d, M. Sharif Kiyani^d, Tahani A. Alrebdi^e, Javed Rehman^f

^a Key Laboratory of Quantum Materials and Devices of Ministry of Education, School of Physics, Southeast University, Nanjing, 211189, China

^b Laboratory of Applied Chemistry (LCA), University of Biskra, P.O. Box 145, 07000, Biskra, Algeria

^c Department of Physics, Faculty of Science and Engineering, University of Nusa Cendana, Kupang, Indonesia

^d Institute of Physics, Post Graduate Program of Material Science, UFRGS, RS, Brazil

^e Department of Physics, College of Science, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh, 11671, Saudi Arabia

^f State Key Laboratory of Metastable Materials Science and Technology, and School of Materials Science and Engineering, Yanshan University, Qinhuangdao, 066004, China

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ABSTRACT

Perovskite materials are a promising candidate for solid-state hydrogen storage applications. Based on a first-principles study, the newly formed AeZrH₃ (Ae = Li/Na/K/Rb/Cs) perovskite hydrides were investigated using van der Waals interactions within the DFT + D approach. This work systematically studies the acoustic and thermophysical aspects of the hydrogen storage characteristics of AeZrH₃ using the density functional theory (DFT) method. The tolerance factor suggests these materials are in the range of stable cubic structures, and AIMD simulation results further confirm their dynamical stability under finite-temperature conditions. With and without spin-orbit coupling (SOC), electronic results show that all hydrides exhibit metallic behavior with a high contribution of Zr d-state. DFT + U analysis indicated KZrH₃, RbZrH₃, and CsZrH₃ have small band gap energies (0.77, 0.82, and 0.99 eV), and the rest (LiZrH₃, NaZrH₃) are metallic compounds, confirmed by electronic properties. Mulliken charge analysis shows the charge transfer characteristics of these compounds, demonstrating that charges move from the Ae and Zr to the H atoms, which makes covalent bonding. AeZrH₃ hydrides exhibited the d-band center, showing a significant upshift for hydrogen uptake and covalent bonding. The elastic properties revealed that NaZrH₃ is the hardest material among them and has a high Debye temperature and thermal conductivity. This study shows the novel AeZrH₃ compounds, providing valuable insights into hydrogen storage applications.

1. Introduction

The two biggest drivers in scientific research today are the need for a pollution-free environment and the energy crisis, which highlights the need to improve the human condition and the planet's growing population [1,2]. Hydrogen is now being viewed as a viable fuel alternative to the power generation industry due to its high energy density, cleanliness, light weight, zero pollutants, and potential for generation with renewable sources [3]. Hydrogen is also unlimited as an energy source compared to fossil fuels, which are finite for stationary and mobile uses.

The biggest hurdle to putting hydrogen-powered vehicles on the market is developing safe but effective storage methods. The current energy crisis and environmental degradation are going to change the case for the energy crisis, the increasing energy consumption, the depletion of fossil fuels, and the degradation of the environment [4]. Technological advancements have led to increased energy consumption in modern societies, further straining the limited availability of fossil fuels. Currently, fossil fuels remain the primary energy source, but their finite nature necessitates the exploration of alternatives. To address these challenges, the US Department of Energy (DOE) has established

* Corresponding author.

** Corresponding author.

E-mail addresses: malikkashif232@gmail.com (M.K. Masood), rania.charif@univ-biskra.dz (R. Charif).

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standards for hydrogen storage techniques. These standards include specific requirements for gravimetric and volumetric storage capacities, temperature ranges (233 to 333 K), and pressures (up to 10 MPa) to be achieved by 2025 [5]. Given the gravimetric and volumetric minima of 5.5 wt% and 40 gH₂/L, respectively, solid-state energy storage indicates energy stored is higher than opposed of liquids or gases [6,7]. Solid-state is advantageous because the solid nature entails higher energy densities, safer operation, and longer cycle life compared to conventional liquid electrolyte-based systems [8]. For example, solid-state batteries use solid electrolytes and electrodes to store and release energy [9]. That said, solid-state batteries can revolutionize energy storage from consumer electronics to electric vehicles to grid-scale energy storage. Other solid-state methods of energy storage include supercapacitors, which store energy through the physical separation of charges at interfaces between solid materials and electrolytes [6,10]. Researchers are investigating solid-state energy storage materials to improve performance metrics, including energy density, charging rates, and cycle life, to facilitate the adoption of renewable energy and the electrification of transportation.

Recently, perovskite hydrides have received a lot of attention due to their potential applications in the field of science and technology, like energy storage, sensors, supercapacitors, fuel cells, and optoelectronics [11–14]. NaMgH₃ hydrides were experimentally studied and found that during the decomposition reaction, the enthalpy and entropy are changed [15]. Martínez-Coronado et al. [16] experimentally synthesized cubic Na_{1-x}Li_xMgH₃ hydride and obtained a desorption temperature in the range of 413–435 °C. The MMgH₃ hydrides were synthesized by the ball milling method; they found that the RbMgH₃ is stable among other compounds [17]. Xu, N. et al. [18] studied the XTiH₃ perovskite-type hydrides for hydrogen storage applications by using the DFT method. They have obtained the maximum gravimetric capacity, 3.36, wt%, respectively. XAlH₃ hydrides were studied for hydrogen storage applications [19]. They found that NaAlH₃ has ductility and KAlH₃ has brittle materials. The calculated gravimetric capacities for NaAlH₃ and KAlH₃ are 5.40 and 4.19 wt%. The new perovskite hydrides XPtH₃ were investigated by using the DFT method, and determined that the hydrogen storage densities are 1.45, 1.35, 1.26, and 1.06 wt%, respectively [20]. Moreover, different hydrides have been systematically studied for hydrogen storage applications like as LiXH₃ [21], LiXH₃ (X = Ca, Sr, Ba) [22], MgXH₃ (X = Al, Sc, Zr) [23], MgXH₃ (X = Cr, Mn, Fe) [24], XFeH₃ (X = Ca, Sr, Ba) [25], XPtH₃ (X = Li, Na, K, Rb) [20], XCuH₃ (X = Rb, K) [26], XSnH₃ (X = Rb, Cs, Fr) [27], CaXH₃ (X = Cu, Zn) [28], and XCrH₃ (X = Na, K) [29], etc. Beyond perovskite hydrides, several complex hydride families such as A₃BH₆, A₂BH₆, and AFe₃H₈ systems have demonstrated diverse structural frameworks and hydrogen storage capacities ranging from moderate to high gravimetric percentages [30, 31]. Besides this, we were interested in investigating zirconium-based hydrides for hydrogen storage applications and transportation. Moreover, they absorb a significant amount of hydrogen, forming zirconium hydrides (ZrH_x), have an excellent capacity to absorb and desorb hydrogen, good kinetics reversibly, and are chemically stable, as well as the d-band center has a significant role in hydrogen uptake and boosting the storage capacity [32].

These are novel perovskite hydrides, various Zr-based has been investigated for hydrogen [33,34] but up to date, theoretical (DFT) and experimental studies on the acoustic, thermophysical aspects, d-band center interaction, Spin-orbit coupling (SOC), and DFT + U approaches of AeZrH₃ (Ae = Li/Na/K/Rb/Cs) for hydrogen storage applications have not been carried out. Therefore, the DFT study was employed for the aforementioned characteristics of AeZrH₃ lightweight perovskite hydrides. The findings were meticulously analyzed and discussed to provide a deeper understanding of the physical characteristics of these materials as promising candidates for solid-state hydrogen storage. This work is structured as follows: Section 2: details the computational methods and procedures used in the study. Section 3: presents and discusses the results related to physical properties. Section 4: d-band center

role and hydrogen storage application, and finally, Section 5: summarizes our computational findings.

2. Computational Methodology

The computational results were conducted on the CASTEP code based on DFT studies [35]. The van der Waals dispersion corrections were implemented by the DFT-D method of the Grimme type and were combined with the GGA-PBE exchange-correlation functional [36]. The core-electron interactions were described by a projector-augmented wave technique [37]. The wave functions were expanded by a plane-wave basis set with a cutoff energy of 680 eV. The Brillouin zone integrations were generated from a Monkhorst-Pack k-point mesh of 8 × 8 × 8 (the spacing of the distance was 0.031 Å⁻¹). After the convergence, the average results of 0.5 × 10⁻⁶ eV for electronic total energy, 0.1 × 10⁻¹ eV/Å for atomic forces, and 0.01 GPa pressure were obtained for all materials at room temperature. Electronic valence shells of the individual atoms that compose the actual hydrides are Li (2s¹), Na (3s¹), K (4s¹), Rb (5s¹), Cs (6s¹), Zr (4 d² 5s²), and H (1s¹). The electronic properties were calculated using a high-dense k-point 12 × 12 × 12 for better electronic density of states. The total convergence energy is 0.5 × 10⁻⁶ eV was achieved. However, we performed the spin-orbit coupling (SOC) calculation of the electronic band structure by using the norm-conserving pseudopotential [38]. Furthermore, the DFT + U method was employed for these materials for the better visualization of valence and conduction bands. The U = 4 eV value was set for the Zr-d state, which was obtained from the literature [39]. The mechanical properties were obtained from the optimized geometry based on a stress-strain method, with a maximum strain amplitude is 0.003. The optical properties were calculated using the k-point mesh 12 × 12 × 12 with the optimized GGA-PBE structure. The optical dielectric functions, refractive index, reflectivity, and adsorption were calculated by eq- (1-5) [14].

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \quad (1)$$

$$n(\omega) = \frac{1}{\sqrt{2}} \left[(\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega))^{1/2} + \varepsilon_1(\omega) \right]^{1/2} \quad (2)$$

$$k(\omega) = \frac{1}{\sqrt{2}} \left[(\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega))^{1/2} - \varepsilon_1(\omega) \right]^{1/2} \quad (3)$$

$$R(\omega) = \frac{(n-1)^2 + k^2}{(n+1)^2 + k^2} \quad (4)$$

$$\alpha(\omega) = \sqrt{2} \omega \left[(\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega))^{1/2} - \varepsilon_1(\omega) \right]^{1/2} \quad (5)$$

Additionally, X-ray diffraction (XRD) patterns were calculated from the relaxation structure and viewed on VESTA software using the wavelength of Cu Kα 1.540 Å [3]. Furthermore, Ab initio molecular dynamics (AIMD) simulations using the VASP software were used to investigate the finite-temperature dynamical stability of all AeZrH₃ (Ae = Li, Na, K, Rb, Cs) hydrides [40,41]. A Nosé–Hoover thermostat was used to maintain the temperature at 300 K while the simulations were run in the canonical NVT ensemble. To provide sufficient sampling of thermal fluctuations, each system was developed for 5 ps (5000 fs) using a time step of 1 fs.

3. Results and discussions

3.1. Structural and materials stability properties

Fig. 1 illustrates the cubic crystal structure of AeZrH₃ perovskite hydride, with the space group number being the *Pm* $\bar{3}$ *m* (#221) [19], and the atomic coordinates of Ae, Zr, and H are 1a (0, 0, 0), 1b (1/2, 1/2, 1/2), and 1c (0, 1/2, 1/2), respectively. Figure S1 shows all the

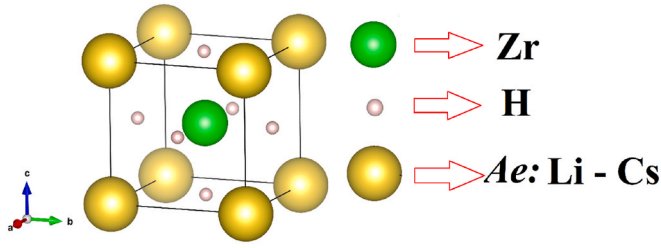


Fig. 1. The systematic cubic crystal structure of AeZrH₃ materials.

optimized structures, and the obtained lattice constants are 3.84, 3.91, 4.05, 4.13, 4.34 Å, and along the volume are 56.62, 59.77, 66.43, 70.44, 81.74 Å³ of LiZrH₃, NaZrH₃, KZrH₃, RbZrH₃, and CsZrH₃, respectively (Table 1). These results demonstrate that increasing the atomic number and valence electrons of materials has increased, as well as varying ionic radii (Cs > Rb > K > Na > Li). The CsZrH₃ has the highest lattice constants among others due to its high volume and ionic radii differences [42]. Moreover, our hydrides lattice parameters results are reasonable to the reported cubic materials KZrH₃ (4.16 Å) [33], LiZrH₃ (3.89 Å) [33], LiTiH₃ (4.42 Å) [43], NaZrH₃ (3.97 Å) [44], CsZrH₃ (4.32 Å) [44], KTiH₃ (3.99 Å) [42], LiSnH₃ (4.10 Å) [45] RbSnH₃ (4.40 Å) [27], CsCoH₃ (3.98 Å) [46], CsZnH₃ (4.02 Å) [46] KCuH₃ (3.73 Å) [26] NaAlH₃ (3.79 Å) [19] and KAlH₃ (3.94 Å) [19]. Furthermore, using the DFT + U approximation, the size of these hydrides is slightly higher than without this function due to an increase in the coulomb repulsion among the localized d-state of Zr. This is an enhanced electronic location and reduced bonding effect.

Additionally, the simulated XRD pattern of the studied materials was obtained from the optimized AeZrH₃ (Ae = Li/Na/K/Rb/Cs) hydride geometries, displayed in Fig. 2. The computed Miller indices are (100), (110), (111), (200), (210), and (211), respectively, for all these samples. The results demonstrate that when the atomic number increases from LiZrH₃ to CsZrH₃, the angle (2 theta) slightly shifts to lower values. However, the main peak (110) from LiZrH₃ to CsZrH₃ shift is clearly noticed with 2 theta angle such as 32.95°, 31.39°, 30.32°, 29.72°, 29.04°, and the varying inter-planar distance (d) 2.715 Å, 2.846 Å, 2.946 Å, 3.003 Å, 3.072 Å, respectively. The ionic sizes increase when moving down any column of the periodic table as a result of the increasing principal quantum number of the outermost electrons. Moreover, the increase in the inter-planar distance, lattice constants, and volume of AeZrH₃ hydrides follows a specific trend (Table 1) [33, 44].

A compound is considered superhard if its bulk modulus exceeds 40 GPa [3]. In this context, the AeZrH₃ hydride bulk moduli exceed 40 GPa, meeting the rigidity criterion. The computed bulk moduli are 76.68, 65.26, 66.58, 74.22, and 41.52 GPa, respectively. Our results indicated that the studied AeZrH₃ materials are hard. It was found that LiZrH₃ is the hardest material of all the studied materials. In addition, our findings are consistent with reported values for other materials, such as NaVH₃ 54.15 GPa [47], RbPtH₃ 74.71 GPa [48], KTiH₃ 44.94 GPa [42], and MgAlH₃ 60.99 GPa [3], respectively. Interestingly, our results are analogous to those reported materials.

To investigate the thermodynamic stability of current materials,

Table 1

The calculated volume (V in Å³), lattice constants (a = b = c in Å), final energy (E_{final} in eV), cohesive energy (ΔE_{coh} in eV/atom), and formation energy (ΔE_f in eV/atom) of AeZrH₃ materials.

Material	V	a = b = c	E _{final}	ΔE _{coh}	ΔE _f
LiZrH ₃	56.62	3.84	-1519.25	-3.09	-0.57
NaZrH ₃	59.77	3.91	-2634.01	-2.98	-0.49
KZrH ₃	66.43	4.05	-2107.62	-2.86	-0.21
RbZrH ₃	70.44	4.13	-1987.34	-2.75	-0.15
CsZrH ₃	81.74	4.34	-1877.46	-2.66	-0.11

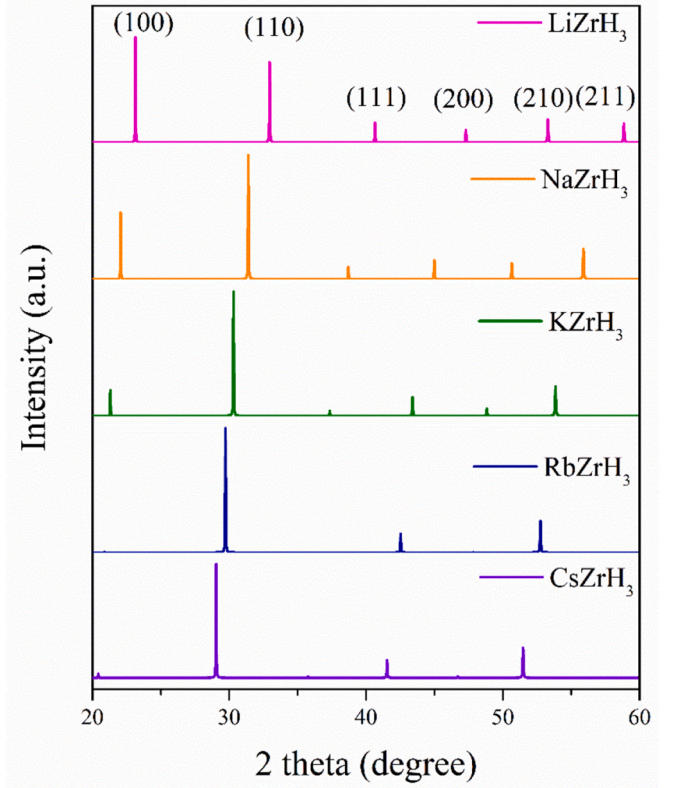


Fig. 2. The X-ray diffraction (XRD) pattern of AeZrH₃ materials.

their formation energy (ΔE_f) was calculated. Materials with negative formation energy are considered thermodynamically stable. The ΔE_f for AeZrH₃ hydrides was calculated in the present work by using eq-(6) [49].

$$\Delta E_f = \left(E_{AeZrH_3}^{total} - E_{Ae}^{solid} - E_{Zr}^{solid} - \frac{3}{2} \times E_{H_2}^{gas} \right) / N \quad (6)$$

Where $E_{AeZrH_3}^{total}$, E_{Zr}^{solid} , E_{Ae}^{solid} , and $E_{H_2}^{gas}$ represent the total ground state energies of the AeZrH₃ compound, Zr atom in solid state, Ae atom in solid state, and H₂ atom in gas state, respectively, while N is the total number of atoms in the system. The calculated formation energies are -0.57, -0.49, -0.21, -0.15, and -0.11 eV/atom of LiZrH₃, NaZrH₃, KZrH₃, RbZrH₃, and CsZrH₃, respectively. Interestingly, all these materials are thermodynamically stable, while the LiZrH₃ compound is suggested to be predominantly more stable among them. However, our results are comparable to other reported materials: KCuH₃, RbCuH₃, NaTiH₃, NaCuH₃, and CsSnH₃ are -0.18, -0.07, -0.85, -0.75, and -0.09 eV/atom [26,27,50].

Additionally, cohesive energy (E_{coh}) was computed for the current AeZrH₃ hydride by using eq-(7) to investigate how strongly these atoms bind with the Zr-based structure and helps determine the stability of AeZrH₃ material. Where the E_{AeZrH₃} is the total energy of the system, E_{Zr}, E_{Ae}, and E_H are the energy of isolated atoms, and N is the total number of atoms, respectively [51].

$$E_{coh} = (E_{AeZrH_3}^{total} - E_{Zr}^{atom} - E_{Ae}^{atom} - 3 \times E_H^{atom}) / N \quad (7)$$

The calculated cohesive energies of LiZrH₃, NaZrH₃, KZrH₃, RbZrH₃, and CsZrH₃ are -3.09, -2.98, -2.86, -2.75, and -2.66 eV/atom, respectively. These results indicate that these atoms are strongly bound within the crystal lattice of the hydrides. Furthermore, the calculated final total energies of these materials are very high, such as -1519.25 eV (LiZrH₃), -2634.01 eV (NaZrH₃), -2107.62 eV (KZrH₃), -1987.34 eV (RbZrH₃), and -1877.46 eV (CsZrH₃), respectively. This indicates these

materials are stable due to more negative ground state energy. Moreover, it has been concluded that the $AeZrH_3$ compounds are all thermodynamically stable and suitable for experimental synthesis as well as further investigation for hydrogen storage applications.

Ab initio molecular dynamics (AIMD) simulations were carried out at 300 K for 5 ps using a canonical (NVT) ensemble to assess the finite-temperature dynamic stability of the Zr-based hydrides. The time-dependent total energy profiles and snapshots of the initial and final atomic configurations for $LiZrH_3$, $NaZrH_3$, $KZrH_3$, $RbZrH_3$, and $CsZrH_3$ are shown in Fig. 3(a–e). The quick thermal equilibration causes the total energy to sharply fall in all cases over the first few hundred femtoseconds. Following this, the systems attain a stable plateau with slight thermal variations. No bond breakage, structural disintegration, and large-amplitude atomic displacements were noted during the simulation. The Zr-H framework maintains its perovskite-like connectivity, and the A-site cations (Li, Na, K, Rb, and Cs) stay in their equilibrium locations without diffusing out of the lattice. The final configurations further demonstrate the hydrides' resilience to thermal excitation by remaining in line with the predicted deformed perovskite structure. These findings unequivocally show that all $AeZrH_3$ compounds are thermally stable at ambient temperature, demonstrating their potential for real-world hydrogen storage applications.

3.2. Mechanically elastic constant properties

Mechanical properties are essential for investigating the materials to determine their mechanical stability [52]. After confirming these factors, they further utilized these materials for their desired applications and devices. Elastic constants allow for the examination of these properties using the Born-Huang mechanical stability criteria concerning the materials. Our $AeZrH_3$ hydride is cubic phase, and this phase has three elastic constants, C_{11} , C_{12} , and C_{44} , respectively. After the calculations, the obtained elastic constant results meet with Born-Huang criteria [29]. Born-Huang mechanical stability equation is: $C_{11} - C_{12} > 0$, $C_{11} + 2C_{12} > 0$, $C_{11} > 0$, and $C_{44} > 0$. Thus, these materials strictly follow the rules, and they exhibit mechanical stability and agree with reported

Zr-based hydrides [33,44,53]. Furthermore, different parameters are calculated by using these elastic constants. Our results demonstrated that the C_{11} values are higher than the C_{12} and C_{44} elastic constants. It means that C_{11} is highly interactive along the axis due to the strong adhesion force along that axis, as well as enhancing the resistance to distortion (Presented in Table 2). Moreover, understanding a compounds mechanical factors and its characteristics, different parameters (stability, anisotropy, ductility, and brittleness), are computed through the application of elastic parameters [54].

Fig. 4 shows the three different moduli of current $AeZrH_3$ materials. These moduli (bulk, shear, and Young) are computed from the elastic constants of these materials. However, the higher the magnitude of

Table 2

The calculated elastic constants (C_{ij}), Pugh's ratio (G/B), Vickers hardness (H_V), anisotropic factor (A), acoustic impedance (Z), radiation factor (I), Debye temperature (θ_D), melting temperature (T_m), Thermal expansion coefficient (α), heat capacity per unit volume (ρC_p), wavelength of the dominant phonon (λ_{dom}), Grüneisen parameter (γ), factor $A(\gamma)$, minimum thermal conductivity (k_{min}), and lattice thermal conductivity (k_{ph}) of $AeZrH_3$ materials.

Parameter	LiZrH ₃	NaZrH ₃	KZrH ₃	RbZrH ₃	CsZrH ₃
C_{11} (GPa)	147.72	127.65	95.53	72.24	55.50
C_{12} (GPa)	17.19	22.42	27.97	33.56	32.86
C_{44} (GPa)	18.66	27.11	32.94	29.34	20.04
G/B	0.52	0.62	0.65	0.53	0.39
H_V (GPa)	7.65	9.38	9.08	6.12	3.27
A	0.28	0.52	0.97	1.52	1.77
Z ($kg\ m^{-2}\ s^{-1}$)	9.69	10.75	10.49	10.25	8.56
I ($m^4\ kg^{-1}\ s^{-1}$)	1.10	1.01	0.95	0.57	0.40
θ_D (K)	482.64	476.52	440.22	332.12	244.36
T_{melt} (K)	911.72	892.81	851.39	827.54	791.80
$\alpha \times 10^{-5}$ (1/K)	5.04	4.51	4.81	6.44	1.00
$\rho C_p \times 10^6$ ($J\ m^{-3}\ K^{-1}$)	3.65	3.46	3.09	2.94	2.53
$\lambda_{dom} \times 10^{-10}$ (m)	1.52	1.53	1.47	1.13	0.87
γ	1.64	1.47	1.41	1.62	1.94
$A(\gamma) \times 10^7$	3.14	3.21	3.23	3.15	3.05
k_{min} ($W\ m^{-1}\ K^{-1}$)	0.99	0.96	0.86	0.63	0.44
k_{ph} ($W\ m^{-1}\ K^{-1}$)	6.80	9.79	10.06	4.37	1.55

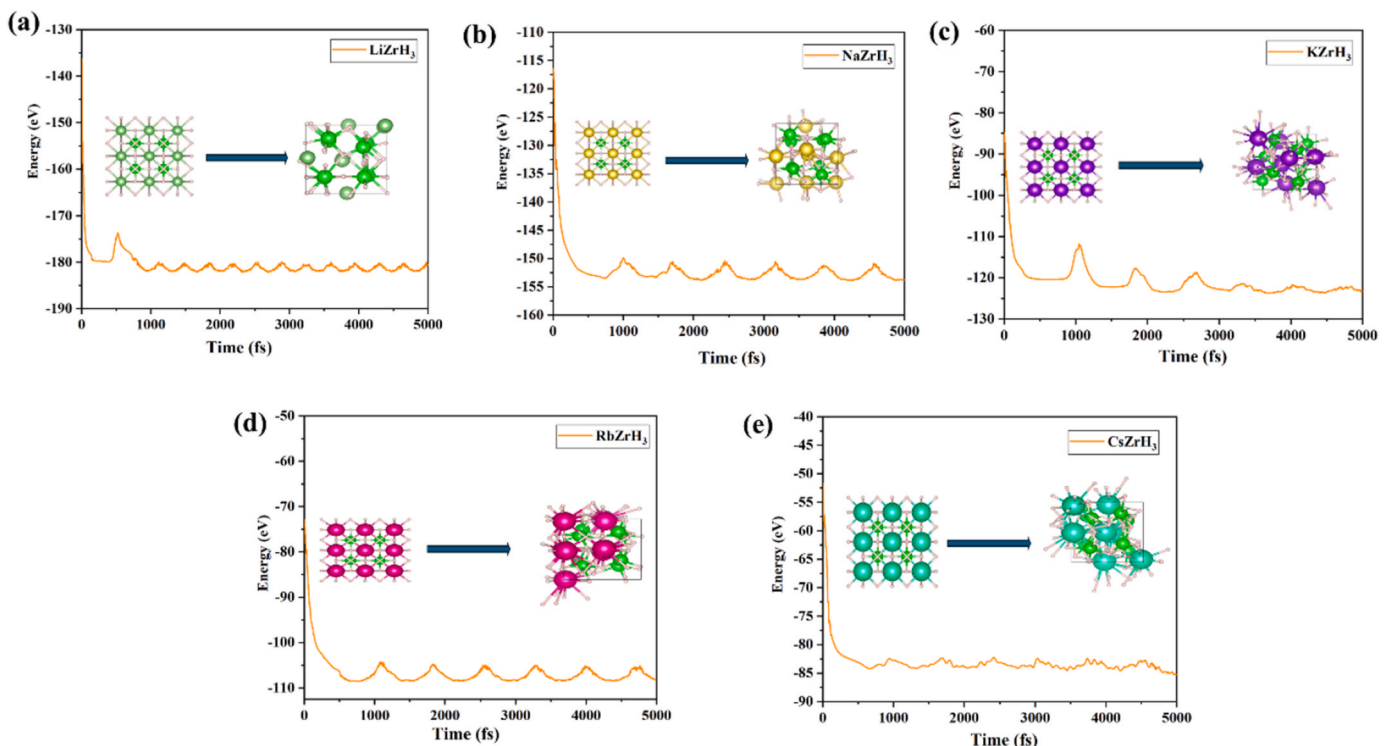


Fig. 3. Ab initio molecular dynamics (AIMD) simulations for (a) $LiZrH_3$, (b) $NaZrH_3$, (c) $KZrH_3$, (d) $RbZrH_3$, and (e) $CsZrH_3$ performed at 300 K for 5 ps.

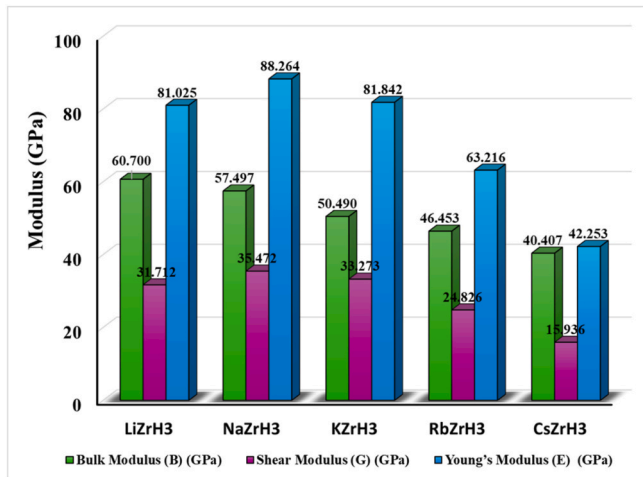


Fig. 4. The plotted elastic modulus of AeZrH₃ hydrides.

moduli, have stronger the ability to withstand shear deformation, tensile and uniform compression, etc. However, the greater the bulk modulus, the higher the resistance of materials, respectively [32]. Our results indicated that LiZrH₃ bulk modulus is greater than that of other hydrides, which exhibits high resistance. The general relation for the shear modulus is the average of both moduli (Reuss and Voigt), $G = \frac{G_V + G_R}{2}$. Fig. 4 demonstrates that NaZrH₃ has a higher shear modulus among the other hydrides, which suggests that NaZrH₃ is more resistant to shear deformation. Through contrasting these ternary hydrides, NaZrH₃ is expected to have a significantly more rigid structure than other hydrides. Moreover, the B value is higher than G of AeZrH₃ materials, which suggests that the mechanical strength of these perovskites is highly constrained by the shear deformation [55]. Another important parameter evaluated for these materials is the Young's modulus, defined as the ratio of tensile stress to tensile strain. This quantity reflects the stiffness of the material and strongly influences its mechanical response under external loading. The Young Modulus expression is $E = \frac{9BG}{3B+G}$ [39], where the B and G are the bulk and shear modulus. It is noticed that NaZrH₃ is the high-stiffness hydride among other hydrides, presented in Fig. 4. Furthermore, the acoustic mode is taken into account for the transformation of thermal behavior due to the group velocity. The group velocity is affected by the mass density and Young's modulus. The thermal conductivity has a relationship with the Young's modulus. Although these are highly impactful factors for materials in different applications like acoustics, their terms also depend on the speed of sound and bulk modulus, respectively. Furthermore, the Young's modulus is used to validate the materials for the thermal barrier coating, ability, and design by using the thermal shock resistance. It is an inverse relationship with Young's modulus. Our results suggested that the AeZrH₃ is suitable for thermal barrier coating applications.

In order to calculate the hardness of materials, which is essential for these hydrides for their applications. The Vicker hardness expression is:

$$H_v = 2 \left(\left(\frac{G}{B} \right)^2 \times G \right)^{0.585} - 3. \text{ Among these materials NaZrH}_3 \text{ is predicted to have the highest } H_v \text{ value. Thus, it is recommended that it is relatively hard material compared to other studied hydrides. However, CsZrH}_3 \text{ is a soft material due to low } H_v \text{ value. However, low even the hardness threshold of fall down of 40 GPa [56]. The machinability index } (\mu_M = \frac{B}{C_{44}}) \text{ provides the information of materials reliability for their effectiveness measurement. Moreover, it is beneficial for quantifying the plasticity and lubricating characteristics of a substance. Our computed results are 3.25, 2.12, 1.53, 1.58, and 2.02 for LiZrH}_3, \text{ NaZrH}_3, \text{ KZrH}_3, \text{ RbZrH}_3, \text{ and CsZrH}_3 \text{ hydrides, respectively. These results show LiZrH}_3 \text{ has highest machinability among the others due to its high value, while}$$

NaZrH₃ and CsZrH₃ are the second-highest materials after KZrH₃ and RbZrH₃ hydrides. Furthermore, these results suggested that AeZrH₃ hydride is suitable for the reliability of their effectiveness measurement and equally appropriate for use in manufactured machines.

The selection of materials for their nature of the ductility (or brittle) characteristics of the AeZrH₃ perovskite hydrides, this factor can be determined by using Poisson's ratio $\nu = \frac{3B-2G}{2(3B+G)}$ and Pugh's moduli (B/G). In literature [55], where the Poisson's (ν) value is lower than 0.26 and Pugh's ratio below 1.75, then it is considered brittle; otherwise, higher values of these indicate a ductile nature. Based on these parameters, the NaZrH₃ and KZrH₃ compounds are brittle, while LiZrH₃, RbZrH₃, and CsZrH₃ compounds are ductile in nature, as shown in Fig. 5. Moreover, the ductile behavior of hydride is highly prevailed for hydrogen storage applications and other features as well. The chemical bonding plays a remarkable role in different characteristics of materials, such as mechanical, chemical, and thermal. Furthermore, the Poisson's ratio (ν) and G/B ratio give the features of the chemical bonding of hydrides; if the Poisson's ratio is within the range of 0.25 and 0.1, while the G/B ratio value is ~ 0.6 and 1.1. Then the compounds are dominated by ionic bonding or covalent bonding characteristics [3]. However, the studied hydrides ν is greater than 0.1 and around ~ 0.25 , which suggests the ionic bonding nature between the atoms. while the G/B computed values are 0.52, 0.62, 0.65, 0.53, and 0.39 for LiZrH₃, NaZrH₃, KZrH₃, RbZrH₃, and CsZrH₃, respectively (Table 2). These results highly predict ionic bonding between the atoms, consistent with the Poisson ratio.

Additionally, the anisotropic factor is highly dominated by the materials technologies [57]. The anisotropic of materials can be computed by the: $A = \frac{2C_{44}}{C_{11} - C_{12}}$. The current AeZrH₃ materials' anisotropic factor is computed and listed in Table 2. Generally, if the anisotropic factor value is 1, while less than or greater than 1, the hydrides are dominated by an isotropic or anisotropic nature [58]. Thus, our compounds follow isotropic trends due to $A \neq 1$: such as 0.28 (LiZrH₃), 0.51 (NaZrH₃), 0.97 (KZrH₃), 1.52 (RbZrH₃), and 1.77 (CsZrH₃), respectively. Furthermore, when $A \neq 1$, the material responds differently along various crystallographic directions, which affects how it deforms during synthesis and processing. Such anisotropy can cause uneven grain growth and residual stresses, making microstructure control more challenging. During hydrogen cycling, direction-dependent expansion may localize strains and promote crack formation. Overall, higher anisotropies can reduce mechanical stability and durability in storage environments. Besides this, the anisotropic elastic properties have been visualized with 2D and 3D directional dependent by employing the ELATE database [59].

Figure S(2-6) presents the 2D and 3D plots of the elastic properties of AeZrH₃ perovskite hydrides, including direction-dependent parameters

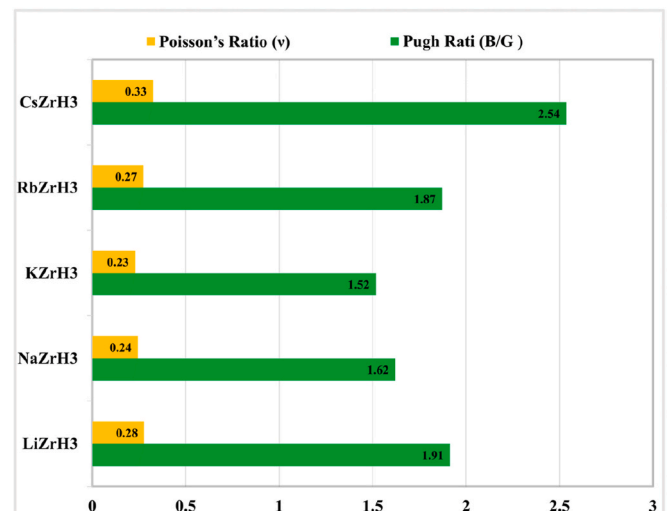


Fig. 5. The plotted Poisson ratio and Pugh ratio of AeZrH₃ hydride.

such as Young's modulus (E), linear compressibility (β), shear modulus (G), and Poisson's ratio (ν). The corresponding minimum and maximum values of these moduli are obtained from the database, summarized in Table 3. A clear trend is observed where both Young's modulus and shear modulus systematically decrease from Li to Cs, indicating a reduction in stiffness across the series. Among these compounds, LiZrH₃ exhibits the highest stiffness. In contrast, the linear compressibility increases with the atomic mass of the Ae-site element, suggesting that heavier atoms contribute to a more compressible structure. Additionally, the nearly spherical shapes of the 3D plots suggest isotropic elastic behavior for all AeZrH₃ hydrides. The Poisson's ratio values, ranging from 0.04 to 0.64, indicate a mix of brittle and ductile characteristics, also pointing toward predominantly ionic bonding in these materials. Furthermore, the relationship between the structural and mechanical properties was analyzed as Li is progressively substituted by heavier alkali metals (Na, K, Rb, and Cs) at the Ae site in AeZrH₃ hydrides. As illustrated in Fig. 6, the lattice parameter "a" and the interplanar distance d(110) show a gradual increase from Li to Cs atom, reflecting the larger ionic radii of the heavier alkali-metal cations. At the same time, the formation energy becomes progressively less negative along the series, indicating a slight reduction in structural stability with increasing atomic size. In contrast to the structural parameters, the mechanical properties exhibit a decreasing trend. The elastic moduli, including the bulk modulus "B", Young's modulus "E", and shear modulus "G", gradually decrease from LiZrH₃ to CsZrH₃. This behavior can be attributed to lattice expansion and the resulting weakening of interatomic bonding. Consequently, the heavier AeZrH₃ compounds exhibit larger lattice volumes but reduced bonding strength and mechanical stiffness.

3.3. Thermophysical properties

The prediction of hydrogen storage capacity in metal hydrides based on their thermodynamic properties plays a vital role in the development of efficient hydrogen storage applications [60]. Key thermodynamic parameters offer insights into the energy required for hydride formation and the stability of the hydrides under varying conditions [61]. Thermal properties of materials are frequently evaluated by different parameters: Debye temperature, melting point, thermal conductivity, and thermal expansion coefficient, which give insight into the thermal stability and thermal character of the system. Debye temperature (θ_D) is universally regarded as one of the most important physical properties in solid-state physics. The θ_D conveys the strength of the bonding between the atoms that make up the material and the energy associated with internal atomic vibrations. Generally, there is a positive relationship between θ_D , phonon thermal conductivity, and θ_D is a dividing line between quantum mechanics and classical mechanics descriptions of phonon behavior. The θ_D is usually calculated based on the elastic constants at low temperatures, characterized by phonon excitations dominated by acoustic phonons. The θ_D is closely associated with various material characteristics, including the melting point and specific heat capacity, and is intrinsically linked to the elastic vibrations of the crystal lattice and the average sound velocity in the material [56]. In this study, the θ_D of the AeZrH₃ hydrides were estimated using the Voigt–Reuss–Hill (VRH) approximation [45], which combines the mechanical moduli to yield an accurate prediction of elastic behavior.

Table 3

The computed minimum and maximum values of Young's modulus, linear compressibility, shear modulus, and Poisson's ratio of AeZrH₃ materials.

Material	Young's modulus (GPa)		Linear compressibility (TPa ⁻¹)		Shear modulus (GPa)		Poisson's ratio	
	$E_{\min}$	$E_{\max}$	$\beta_{\min}$	$\beta_{\max}$	$G_{\min}$	$G_{\max}$	$\nu_{\min}$	$\nu_{\max}$
LiZrH ₃	50.78	1441.3	0.54	0.54	186.63	652.63	0.04	0.62
NaZrH ₃	702.9	1209.5	0.57	0.57	271.13	526.15	0.09	0.44
KZrH ₃	811.75	828.66	0.66	0.66	329.43	337.83	0.22	0.23
RbZrH ₃	509.55	727.07	0.72	0.72	193.43	293.38	0.13	0.42
CsZrH ₃	310.63	515.83	0.83	0.83	113.21	200.37	0.12	0.53

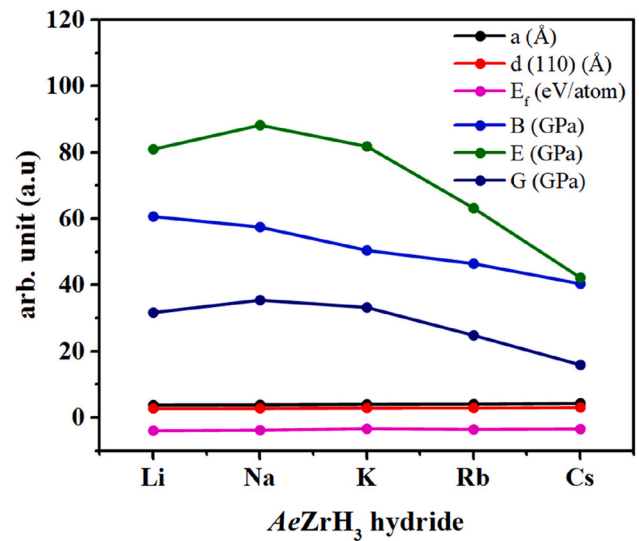


Fig. 6. structural and mechanical trends of AeZrH₃ hydrides, demonstrating how lattice parameters, formation energy, and elastic moduli effected with substitution of Li to Cs atoms.

$$\theta_D = \frac{h}{k_B} \left(\frac{3n N_A \rho}{4\pi M} \right)^{1/3} V_m \quad (8)$$

Table 2 shows the calculated θ_D for the AeZrH₃ hydrides. The results depict that the stiffness follows the trends LiZrH₃ > NaZrH₃ > KZrH₃ > RbZrH₃ > CsZrH₃, respectively. A high value of the θ_D indicates a high thermal conductivity and a high melting temperature (T_{melt}) for the material. Our computed values are not greater than 500 K, even decreasing toward room temperature; it may be interesting for the hydrogen applications due to the low θ_D indicating softer lattice vibrations and weaker atomic bonding. Furthermore, this can enhance hydrogen diffusion and improve absorption/desorption kinetics. Another factor is that the T_{melt} of these cubic crystals was computed from eq-(9) for AeZrH₃ hydrides [3].

$$T_m = [553 + 5.911 \times C_{11}] \text{GPa} \pm 300 \text{ K} \quad (9)$$

The calculated melting temperatures for AeZrH₃ compounds are presented in Table 2. In particular, materials with low and high melting temperatures would be good candidates for thermal interface material and thermal barrier coatings. A consistent trend is observed where the melting temperature decreases with an increase in the atomic mass of the Ae-site element, progressing from Li to Cs. Among the studied materials, LiZrH₃ exhibits the highest melting temperature, while CsZrH₃ shows the lowest. This indicates that LiZrH₃ possesses superior thermal stability compared to its heavier counterparts. The AeZrH₃ hydride has high melting temperature compounds, so they can likely be used in harsh thermal environments. Markedly, the melting temperature increases with higher bonding strength and greater cohesive energy. In addition to its high melting point, LiZrH₃ also demonstrates the highest values for θ_D and thermal conductivity, making it a particularly promising candidate in terms of thermodynamic performance.

The thermal expansion coefficient (α) is an important physical property that measures how much a material changes dimensions with temperature. The α is a property that is dependent on an harmonic lattice vibrations and is a critical component to identify if a material is suitable for application to thermal barrier coating. The α is also fundamental in other material properties such as, thermal conductivity, specific heat, entropy, and isothermal compressibility. The α can be calculated from the shear modulus G in the relation is expressed below [62]:

$$\alpha = \frac{1.6 \times 10^{-3}}{G} \quad (10)$$

These values are calculated for AeZrH₃ hydride, are listed in Table 2. Our results indicated that RbZrH₃ is the highest value among them; it is validated that it expands upon heating and contracts upon cooling more than others, while LiZrH₃, NaZrH₃ and KZrH₃ have the same trend but are lower than Rb hydride. Where the Cs hydride shows a very low effect. It also shows that there is an inverse relationship between the thermal expansion coefficient and melting temperature, such that:

$$\alpha = \frac{0.02}{T_{\text{melt}}} \quad (11)$$

Attempting to also extrapolate the volumetric heat capacity (ρC_p), which is a measure related to the amount of heat energy stored per unit volume of material for a 1 K temperature change and calculated with the relationship:

$$\rho C_p = 3k_B/\Omega \quad (12)$$

where Ω is the average volume per atom and $N = 1/\Omega$ is the number of atoms per volume. This parameter is essentially the storage capacity or ability to store thermal energy and how sensitive the material is to temperature changes. Table 2 shows the data computed for the AeZrH₃ volumetric heat capacity, it is indicated that increasing the atomic radii and mass number, the volumetric heat capacity decreases from Li to Cs atoms, respectively. Moreover, quantum lattice vibrations and phonons play an essential role in many physical properties, like electrical conductivity, thermal conductivity, and heat capacity. The dominant phonon wavelength (λ_{dom}) is associated with the wavelength corresponding to the peak of the phonon distribution function. The λ_{dom} was estimated for AeZrH₃ hydride at 300 K using the relationship provided previously. Table 2 shows that the trend LiZrH₃ > NaZrH₃ > KZrH₃ > RbZrH₃ > CsZrH₃, due to having a longer dominant phonon wavelength, signifying the increase in wavelength or the shift of the peak of the phonon distribution towards larger wavelengths.

$$\lambda_{\text{dom}} = \frac{12.566 v_m}{T} \times 10^{-12} \quad (13)$$

The Grüneisen parameter (γ) is an important thermophysical property that depends in general on both the volume and temperature of a material and is related to the strength of phonon–phonon interactions. The γ can provide information about thermal conductivity, thermal expansion, temperature dependence of elastic constants, attenuation of acoustic waves, and the degree of anharmonicity of the atomic bonds in the crystal. In this study, we have evaluated the γ for AeZrH₃ hydride through the following expression:

$$\gamma = \frac{3}{2} \left(\frac{1 + \nu}{2 - 3\nu} \right) \quad (14)$$

where ν represents the Poisson ratio. Thus, the calculated values of γ for AeZrH₃ hydride (Table 2) demonstrate that, attributed to significant anharmonicity, these materials have poor thermal conductivity. Furthermore, high γ means that the thermal attributes of the materials, such as thermal expansion, specific heat, and thermal conductivity, are highly sensitive to temperature and pressure variations. Furthermore, the study of the minimum phonon thermal conductivity of a material at

an elevated temperature is important in seeking and creating materials to use as thermal barrier coating materials. The minimum thermal conductivity of AeZrH₃ hydride is identified as k_{min} and is defined at high temperature using Clarke's formula:

$$k_{\text{min}} = k_B v_m \left(\frac{n N_A \rho}{M} \right)^{2/3} \quad (15)$$

In this equation, ρ is the mass density and M is the molecular weight. Where the N_A , n , k_B , and v_m are the Avogadro number, number of atoms, Boltzmann constant, and average sound speed. When evaluating the thermal insulation capacity of hydrogen storage materials, minimum thermal conductivity (K_{min}) is crucial. It is important to note that low thermal conductivity is highly desired for use in thermoelectric devices and thermal insulators. K_{min} falls from LiZrH₃ (0.99 W/m·K) to CsZrH₃ (0.44 W/m·K) among the Zr-based hydrides under study, indicating the effect of increasing alkali metal mass on phonon scattering and lattice softness. Better heat retention is indicated by lower K_{min} values, as observed in RbZrH₃ and CsZrH₃. This is advantageous during hydrogen desorption. These AeZrH₃ compounds are comparatively very low in minimum thermal conductivity as shown in Table 2. The AeZrH₃ hydride is a good potential candidate for thermal barrier coating materials at ambient pressure values, since their k_{min} values are lower than the critical thermal conductivity of $1.25 \text{ W m}^{-1} \text{ K}^{-1}$ [63]. Therefore, materials with lower K_{min} are thermally beneficial for hydrogen release; however, for practical usage, this needs to be balanced with structural stability and storage capacity. They are thermally advantageous for storing hydrogen because of their decreased phonon transit, which aids in heat retention within the system.

$$k_{\text{ph}} = A(\gamma) \frac{M_{\text{av}} \theta_D^3 \delta}{\gamma^2 n^{2/3} T} \quad (16)$$

$$A(\gamma) = \frac{5.72 \times 10^7 \times 0.849}{2 \times \left(1 - \frac{0.514}{\gamma} + \frac{0.228}{\gamma^2} \right)} \quad (17)$$

The calculated lattice thermal conductivity values at room temperature are summarized in Table 2. The phonon thermal conductivity (K_{ph}) of LiZrH₃, NaZrH₃, KZrH₃, RbZrH₃, and CsZrH₃ at 300 K are 6.8, 9.79, 10.06, 4.37, and $1.55 \text{ W m}^{-1} \text{ K}^{-1}$, respectively. These results reveal a noticeable reduction in thermal conductivity when Li is substituted by heavier alkali metals such as Na, K, Rb, and Cs. This trend is consistent with the decrease in Debye temperature, since lower Debye temperatures generally indicate weaker interatomic bonding and reduced phonon transport efficiency. Furthermore, the larger atomic radius of Cs increases the lattice spacing, which enhances phonon scattering and consequently lowers the lattice thermal conductivity. As a result, CsZrH₃ exhibits superior thermal insulation characteristics compared with the other Zr-based hydrides, making it a promising candidate for applications requiring thermal barrier coatings and thermoelectric materials. In contrast, NaZrH₃, which shows relatively higher thermal conductivity, may be more suitable for heat dissipation applications in microelectronic and nanoelectronic devices. Fig. 7 depicts the lattice thermal conductivity of the studied materials; however, the atomic and bonding aspects of AeZrH₃ hydride are related to lattice thermal conductivity. Furthermore, the NaZrH₃ and KZrH₃ show high lattice thermal conductivity as compared to other hydrides due to a stronger bonding nature involving these materials, like covalent bonding. It can minimize the phonon dispersion and improve heat transmission. Although the other materials also show aspects at different temperatures and different k_{ph} values. CsZrH₃ hydride shows very low lattice thermal conductivity among them.

3.4. Acoustic properties

A material's acoustic characteristics fundamentally govern its

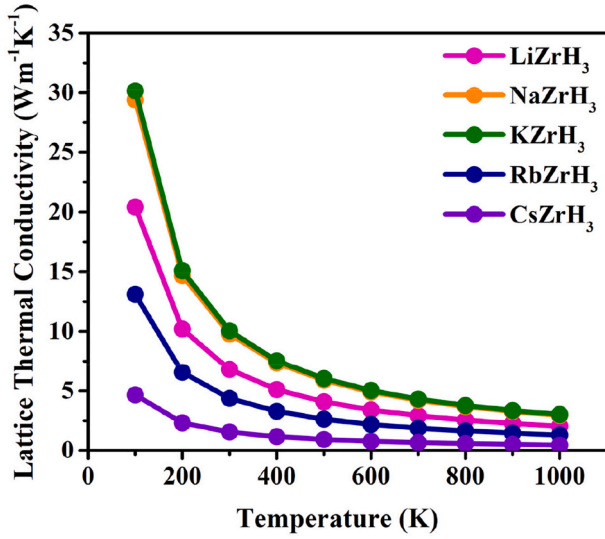


Fig. 7. The typical variation of lattice thermal conductivity as a function of temperature of AeZrH₃ hydride.

thermal and electric conduction [64]. Materials that allow proper management of the transmission of sound should keep in mind noise pollution and thus require materials that can appropriately and efficiently reflect sound waves [65]. To understand the acoustic waves through a material, one must typically look at the 2nd-order elastic constants of a material. The materials studied in this work are described greatly with the key acoustic parameters and some relevant sound information, including sound velocity, sound anisotropy, acoustic impedance, and the sound radiation coefficient.

Acoustic velocity is the speed at which sound travels through a solid material and is, first and foremost, a function of the elastic properties of that same material. Each elastic property relates to a particular speed. e. g, in solids, there are two basic categories: longitudinal velocity (V_l) and transverse velocity (V_t). There is also average sound velocity (V_m), and some crystalline solids have longitudinal, transverse, and average sound velocities based on equations and calculated for AeZrH₃ perovskite hydrides [49]:

$$V_m = \left[\frac{1}{3} \left(\frac{2}{V_t^3} + \frac{1}{V_l^3} \right) \right]^{-\frac{1}{3}} \quad (18)$$

$$V_t = \sqrt{\frac{G}{\rho}} \quad (19)$$

$$V_l = \sqrt{\frac{3B + 4G}{3\rho}} \quad (20)$$

In this equation, the G , B , and ρ represent the shear modulus, bulk modulus, and density of the AeZrH₃ hydride, respectively. These velocities (V_l , V_t , V_m) for the AeZrH₃ hydrides are calculated and presented in Fig. 8. However, these results revealed the trend $\text{LiZrH}_3 > \text{NaZrH}_3 > \text{KZrH}_3 > \text{RbZrH}_3 > \text{CsZrH}_3$, respectively. It is exhibiting the values of V_m , V_l , and V_t are decreased from Li to Cs atom substitution on AeZrH₃ hydrides. It suggests these hydrides are mechanically tougher. which can be attributed to its relatively high Debye temperature. This suggests that AeZrH₃ is a mechanically hard material with superior thermal conductivity. A solid possesses three vibrational modes for each atom: two transverse and one longitudinal. In anisotropic materials, pure longitudinal and transverse wave modes will only occur in certain crystallographic directions. For all other directions, the vibrational modes are classified as quasi-longitudinal and quasi-transverse. There are only certain directions along which a compound with cubic

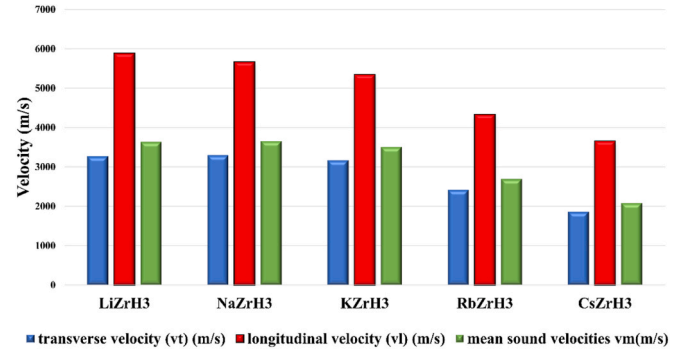


Fig. 8. The plotted velocities (ms^{-1}) of sound for AeZrH₃ hydrides.

symmetry can have pure longitudinal and transverse modes. The single-crystal elastic constants of AeZrH₃ can be used to calculate the acoustic velocities of the material along its three principal directions, as shown in Fig. 9. The velocity of elastic waves within a material depends only on that material's intrinsic properties, which remain unchanged, regardless of the wave's size or the frequency of the wave. Our findings suggested that longitudinal sound velocities have a higher impact on these materials as compared to transverse sound velocities in the direction of (100), (110), and (111), respectively. However, these results revealed that trend for hydride are $\text{LiZrH}_3 > \text{NaZrH}_3 > \text{KZrH}_3 > \text{RbZrH}_3 > \text{CsZrH}_3$, respectively. Furthermore, the sound velocities are significantly dependent on the direction in the AeZrH₃ hydrides, suggesting the possibility that similar anisotropy could also affect their thermal and electrical transport properties.

In the [100] direction:

$$\text{Longitudinal velocity} = v_l = \sqrt{\frac{C_{11}}{\rho}} \quad (21)$$

$$\text{Transverse velocity} = v_t = \sqrt{\frac{C_{44}}{\rho}} \quad (22)$$

In the [110] direction:

$$\text{Longitudinal velocity} = v_l = \sqrt{\frac{C_{11} + C_{12} + 2C_{44}}{2\rho}} \quad (23)$$

$$\text{Transverse velocity first mode} = v_{t1} = \sqrt{\frac{C_{11} - C_{12}}{2\rho}} \quad (24)$$

$$\text{Transverse velocity second mode} = v_{t2} = \sqrt{\frac{C_{44}}{\rho}} \quad (25)$$

In the [111] direction:

$$\text{Longitudinal velocity} = v_l = \sqrt{\frac{C_{11} + 2C_{12} + 4C_{44}}{3\rho}} \quad (26)$$

$$\text{Transverse velocity} = v_t = \sqrt{\frac{C_{11} - C_{12}}{3\rho}} \text{ (degenerate)} \quad (27)$$

Acoustic impedance ($Z = \sqrt{\rho G}$) is an important parameter that affects the transfer of acoustic energy between two materials. The amount of sound that is transmitted and reflected at the interface between two materials is dependent on the difference in acoustic impedances. Overall, when the impedances of each material match closely, sound transmission is more efficient. Table 2 shows that Z values for AeZrH₃ hydride, where NaZrH₃ supports sound propagation more rapidly than others, while the following trend $\text{NaZrH}_3 > \text{KZrH}_3 > \text{RbZrH}_3 > \text{LiZrH}_3 > \text{CsZrH}_3$, respectively, due to the Z values. The sound radiation

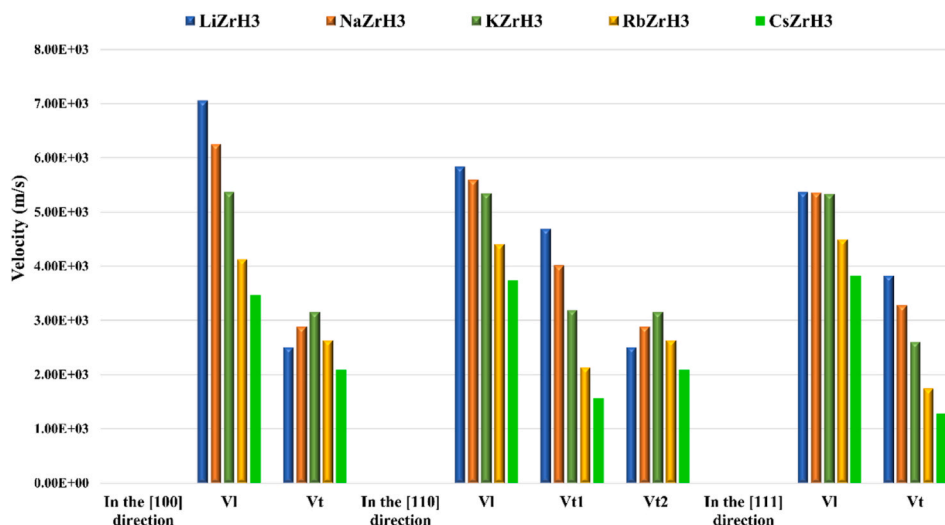


Fig. 9. The plotted Anisotropic sound velocities (ms^{-1}) in different directions (100), (110), and (111) for AeZrH₃ hydride. Where each color represents a different hydride. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

intensity is another critical component of acoustic material design and is important for developing acoustic insulation and soundproofing materials. The sound radiation intensity ($I = \frac{v}{\rho} = \sqrt{\frac{G}{\rho^3}} (m^3 \text{kg}^{-1} \text{s}^{-1})$) is directly proportional to the surface velocity of the material, for a given source. The amount of sound radiation is dependent on the material's stiffness modulus and density; the higher the stiffness and the lower the density, the sounder radiation occurs. However, makers of musical instruments usually select those materials with large radiation factor values to make effective soundboards to promote sound production. Our findings suggested that LiZrH₃ is the highest sound radiation intensity among the others, due to higher stiffness and lower density. However, they are following the same trend as like acoustic directional sound velocities. The other important aspect is the amount of acoustic conversion efficiency, which converts a mechanical vibration into sound energy. The acoustic conversion efficiency value is closely related to the values for the radiation factor. The computed values exhibited the radiation factor of AeZrH₃ hydride, enabling further evaluation of their acoustic properties. Table 2 shows these values decrease as move to high ionic radii atoms from Li to Cs, despite showing the same trend as follows by the sound radiation intensity.

3.5. Band structures and spin-polarized density of states

The electronic band structure of AeZrH₃ was investigated by using the GGA-PBE with and without SOC, and the DFT + U method in the first Brillouin zone, with the high symmetry K points, displayed in Figure 10–12. Clearly, the AeZrH₃ compounds with and without SOC effect demonstrate metallic characteristics as the bands at the E_F facilitate electron transitions between valence (V.B) and conduction bands (C.B), displayed in Fig. 10. This is enhancing rapid charge transfer, and the same behavior is observed in the total density of states (TDOS) of AeZrH₃ (Fig. S7). Although with DFT + U results exhibited, the LiZrH₃ and NaZrH₃ are metallic hydrides, whereas the KZrH₃, RbZrH₃, and CsZrH₃ hydrides have small indirect band gap energies of 0.77, 0.82, and 0.99 eV, respectively, presented in Fig. 11. These materials exhibit semiconductor behavior when applying the DFT + U method. Furthermore, the difference arises mainly from the lattice size and resulting Zr-H bonding. In LiZrH₃ and NaZrH₃, the smaller Ae-site ions create a compact lattice, leading to strong orbital overlap and wide Zr-d bands that still cross the E_F even after applying U = 4 eV, so these hydrides remain metallic. For K, Rb, and Cs, the larger ionic radii expand the lattice, reduce Zr-H overlap, and narrow the Zr-d bandwidth. In this

case, the Hubbard U is sufficient to separate the valence and conduction states, opening a band gap and producing semiconducting behavior. Thus, the metallic and semiconducting nature is controlled by the balance between lattice expansion and Zr-d correlation effects. Additionally, the spin-polarization results are analyzed on the ground state, which validates that the AeZrH₃ hydrides are of an anti-ferromagnetic nature due to spin-up and spin-down being symmetrical (Fig. 10). Furthermore, all hydrides show non-magnetic characteristics.

Metallic nature materials play a significant role in hydrogen storage and transportation applications [66]. This attribute facilitates the movement of hydrogen atoms, thereby improving the kinetics of storage and release. Moreover, our results are in good agreement with the literature: LiZrH₃ (metallic) [33], NaZrH₃ (metallic) [44], KZrH₃ (metallic) [33], CsZrH₃ (metallic) [44], RbNiH₃ (metallic) [48], KPdH₃ (metallic) [67], CaMgH₃ (metallic) [68], and LiPtH₃ (metallic) [20] studied for hydrogen storage application by using DFT method. To gain a deeper understanding of the bonding characteristics between atoms and the role of individual atomic orbitals in electrical conductivity and other electronic transport phenomena, it is essential to analyze both the total (TDOS) and partial density of states (PDOS) of materials. Notably, the electronic contribution to properties such as spin paramagnetic susceptibility and the electronic component of the heat capacity is closely linked to the density of states at the E_F. Fig. S7 display the calculated TDOS for AeZrH₃ as a function of energy (eV) by using the method without DFT + U and SOC approximation. The finite value of TDOS at the E_F indicates the metallic nature of AeZrH₃ hydrides. The computed values in TDOS at the E_F are 0.46, 0.54, 0.56, 0.63, and 1.76 states/eV per unit cell, respectively, for AeZrH₃ (Ae = Li/Na/K/Rb/Cs) hydrides. These results imply that CsZrH₃ may exhibit superior electrical conductivity among them. However, LiZrH₃ appears to possess greater thermodynamic stability. Interestingly, all these hydride shows more conduction bands intersecting the Fermi level; CsZrH₃ has a higher density of states at the E_F. These characteristics influence electrical conductivity in different ways. Overall, AeZrH₃ hydrides are expected to demonstrate good metallic conductivity.

The spin-polarized density of states of the studied materials was employed by the spin-restricted, with and without SOC, the DFT + U method (Figure 13–14). The PDOS demonstrated each state of an atom, which contributes to TDOS. Fig. 13 illustrates that the high contribution of the Zr-4d and H-1s state in V.B, and Ae s-state is low dominant in C.B. However, Zr-4d states play a remarkable role near the E_F in all hydrides. The highly intense peaks are observed, indicating a significant role in TDOS of these materials. These results revealed that strong

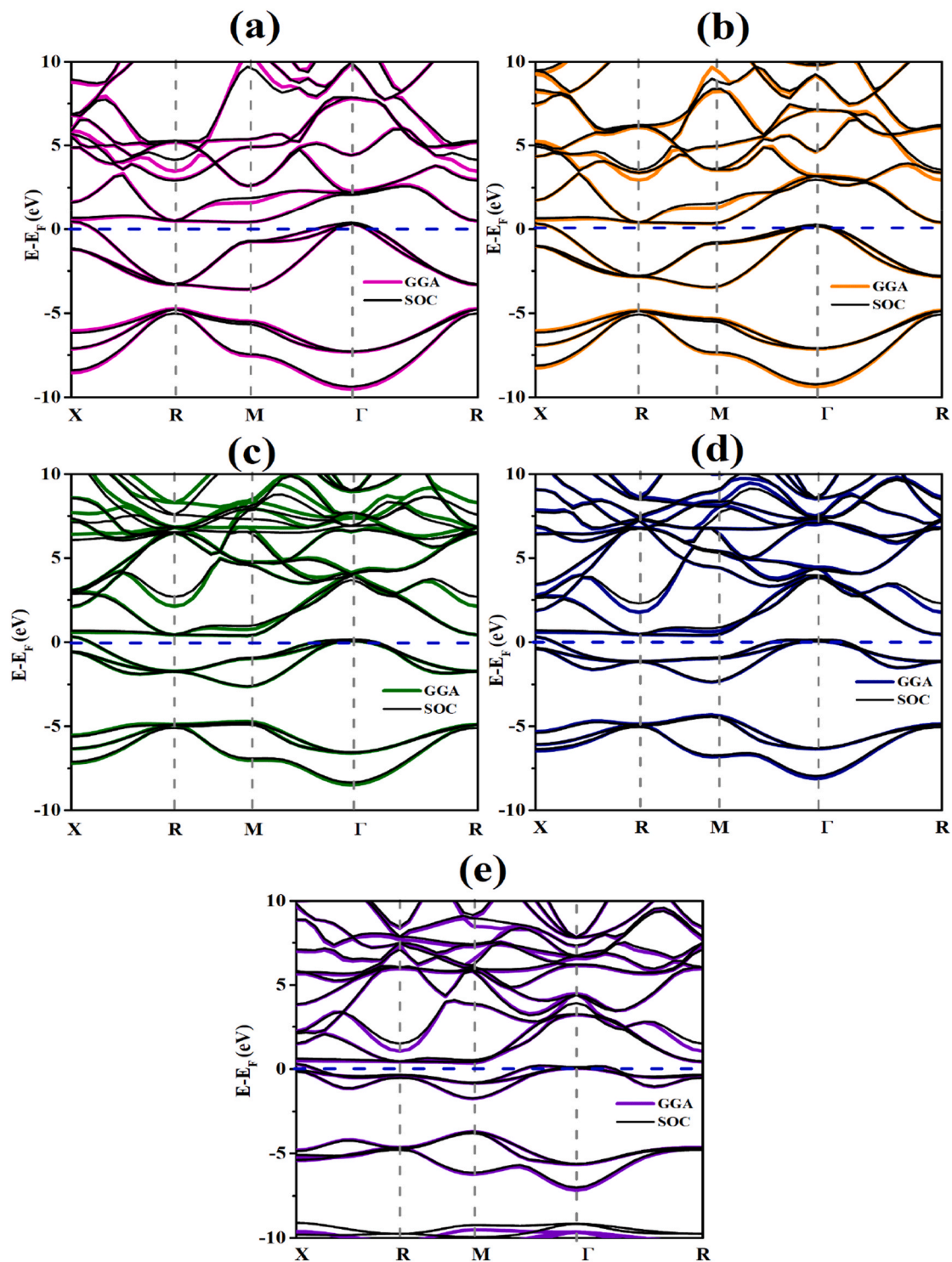


Fig. 10. Electronic band structure of (a) LiZrH₃, (b) NaZrH₃, (c) KZrH₃, (d) RbZrH₃, and (e) CsZrH₃ hydrides, with and without SOC functional.

hybridization between the Zr-d state and Ae-s state emerges from these bonding and anti-bonding peaks, showing a metallic nature in all AeZrH₃ hydrides. Interestingly, metallic hydrides have a promising role in hydrogen storage applications [42,48]. Furthermore, the anti-bonding peaks in LiZrH₃ exhibit distinct features compared to those observed in other hydrides, as shown in Fig. 13(a). When Li is substituted with alkali metals such as Na, K, Rb, and Cs in the AeZrH₃ materials, noticeable variations in the anti-bonding peaks near the E_F are observed (Fig. 13(b-e)). These differences highlight the sensitivity of the electronic structure to chemical substitution. Additionally, a

pronounced dip in the TDOS near E_F, referred to as a pseudogap or quasi-gap is evident [62]. This pseudogap separates bonding and anti-bonding states and is commonly associated with enhanced electronic stability. Its proximity to the E_F also suggests the presence of strong covalent bonding, which often correlates with improved mechanical strength and structural robustness. However, similar contributions from V.B and C.B were noticed in PDOS with SOC and DFT+U case (Fig. 14).

To delve deeper into the charge transfer phenomenon and bonding nature within AeZrH₃ hydrides, we calculated the Mulliken and

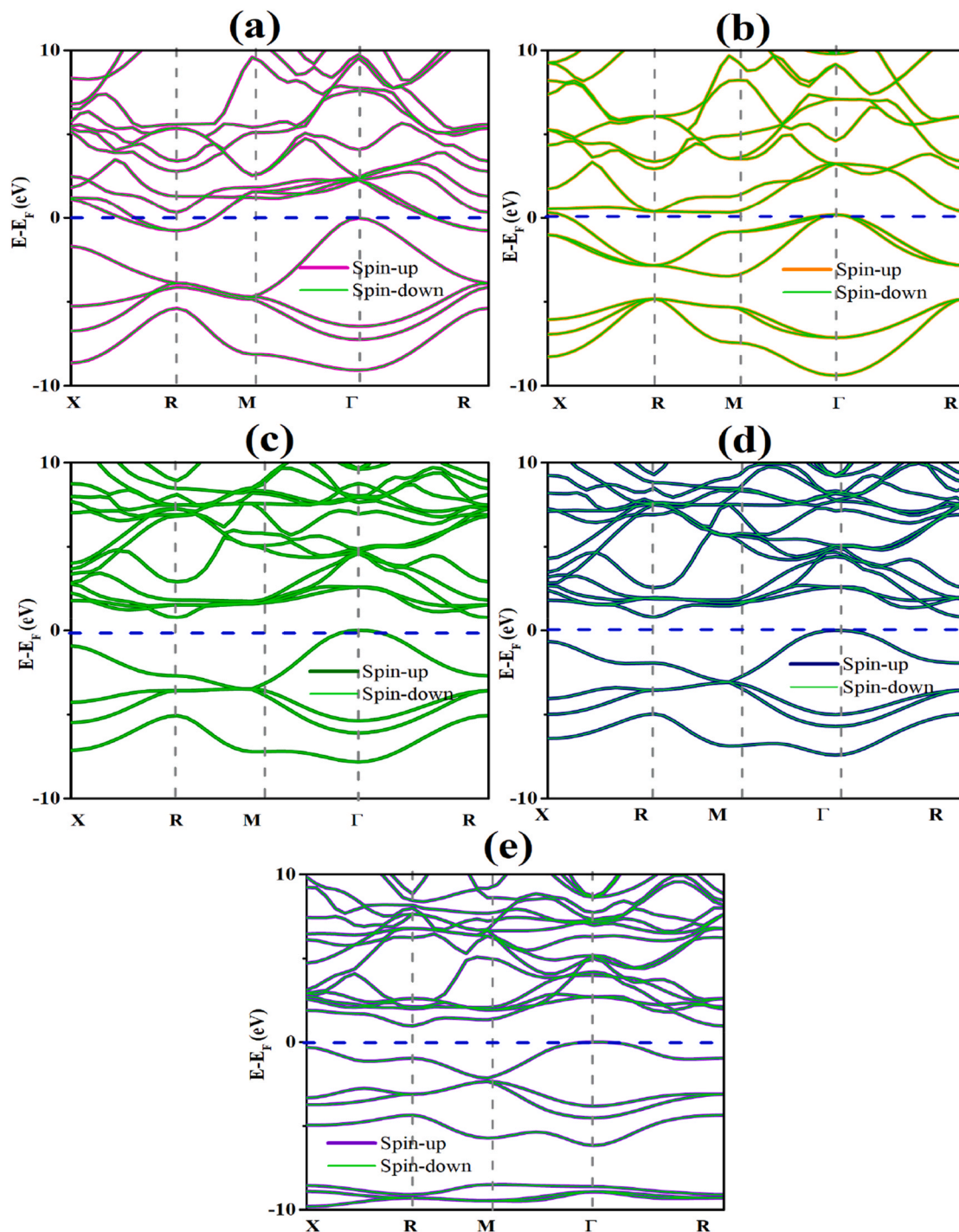


Fig. 11. Spin-polarized electronic band structure of (a) LiZrH₃, (b) NaZrH₃, (c) KZrH₃, (d) RbZrH₃, and (e) CsZrH₃ hydrides, DFT + U.

Hirshfeld's charges and presented them in Table 4 and Table S2. The negative and positive partial charges signify absorbed and released charges, and the sum of the net partial charge is zero. Although Mulliken and Hirshfeld's charges are analogous. Charges analysis results reveal electron transfer from Ae and Zr atoms to H atoms, indicating the formation of an ionic bond between the atoms. The ionic character between Li-H and Zr-H suggests that hydrogen is strongly bound in the lattice. Moreover, formal ionic charges of Ae (+2), Zr (+4), and H (+1/-1) atoms, where the calculated effective Mulliken and Hirshfeld's charges are reasonably lower than the ionic charges of the atoms. Besides this, the bond population was calculated for the current perovskite hydrides, displayed in Table S1. This shows that the increasing atomic masses of

atoms increase the bond length of compounds. Thus, these results are consistent with the structural properties of the AeZrH₃ lattice parameters and XRD results. Additionally, the electron density distribution of AeZrH₃ compounds along the (1 1 0) plane is presented in Fig. 15. However, the blue, green, and red colors represented the ionic, metallic, and covalent bonding of materials, as well as light blue and green-yellow colors indicated metallic-ionic and metallic-covalent bonds between the atoms. These results indicated that H atoms and Zr atoms occupy a higher density than Ae atoms due to high colors, even Zr shows a much higher density because high number of electrons in the valence shell. It is noticed that ionic bonding exists between the Zr-H, H-Ae, and Zr-Ae, respectively, where the supporting conclusion is drawn from Poisson's

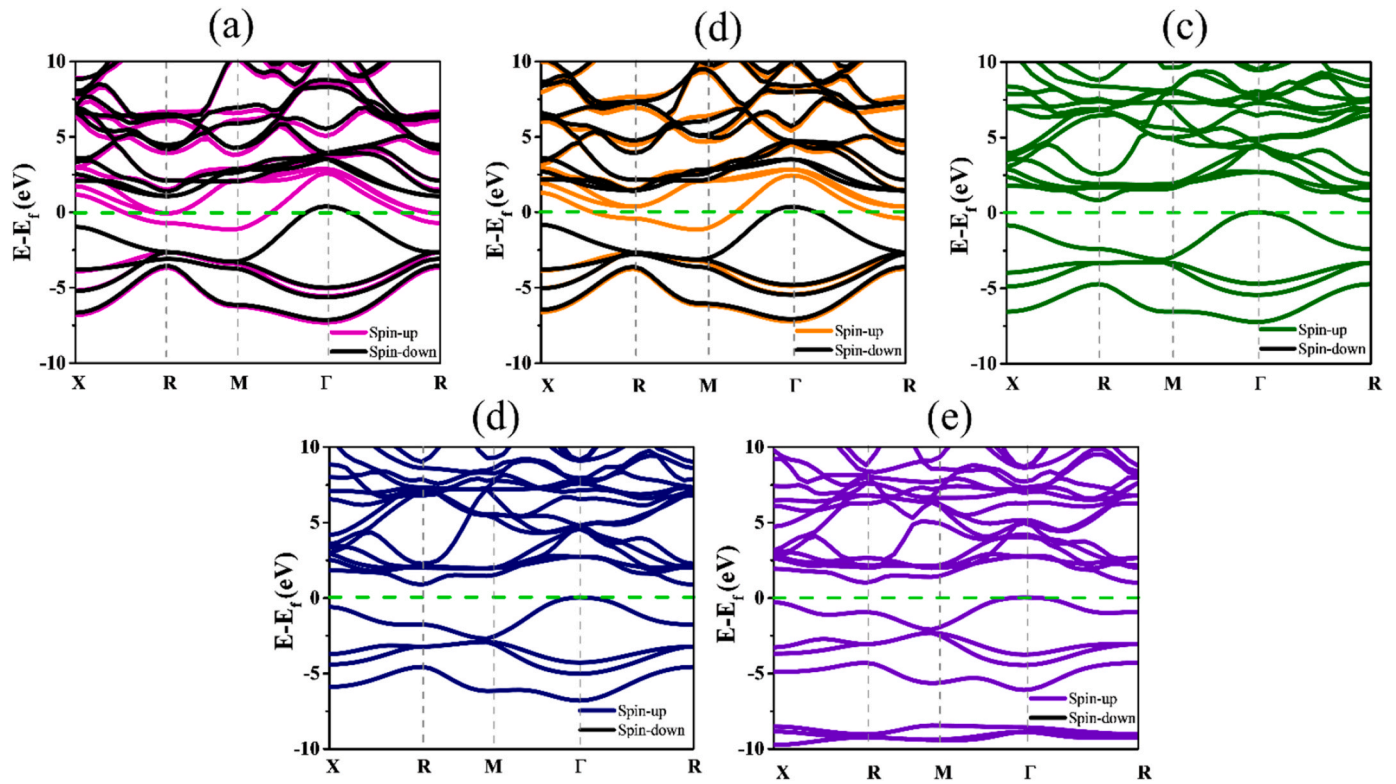


Fig. 12. The plotted bands structure with SOC and DFT + U method, (a) LiZrH₃, (b) NaZrH₃, (c) KZrH₃, (d) RbZrH₃ and (e) CsZrH₃ materials.

and Pugh's ratio analysis regarding the ionic bonding nature of AeZrH₃.

3.6. Optical properties

Understanding the optical properties of materials is paramount for their application in various fields, including optoelectronics and energy storage, as well as for gaining insights into their characteristics through electromagnetic radiation [69]. This section delves into the discussion of AeZrH₃ hydrides under GGA approximation, aiming to assess their optical spectra about electronic characteristics. Various optical parameters are examined, including the dielectric function $\epsilon(\omega)$, refractive index $n(\omega)$, absorption coefficients $\alpha(\omega)$, and conductivity $\sigma(\omega)$, with the incident photon frequencies for AeZrH₃ hydrides. Central to this analysis is the $\epsilon(\omega)$, which plays a pivotal role in comprehending both the optical and electrical behaviors of materials, encompassing their responses to light, EM waves, and electronic transport.

The complex $\epsilon(\omega)$ comprises two crucial components: $\epsilon(\omega) = \epsilon_1(\omega) + i\epsilon_2(\omega)$. Where the $\epsilon_1(\omega)$, representing the real part that elucidates dispersion and polarization phenomena, and $\epsilon_2(\omega)$, denoting the imaginary part that characterizes the absorption properties of materials [70]. Fig. 16 (c) illustrates the optical dielectric function for AeZrH₃ materials. Notably, $\omega = 0$, the $\epsilon_1(0)$ values for AeZrH₃ are 15, 11, 23 for LiZrH₃, NaZrH₃ and RbZrH₃, where the KZrH₃ and CsZrH₃ are higher than 25. While varying the incident photon energy, the significant peak values of $\epsilon_1(\omega)$ component at different energies, which is vary around 5 eV. After this, both are drastically dropped to minimum values of $\epsilon_1(\omega)$, where it has to oscillate and become parallel around 15 eV. Notably, CsZrH₃ demonstrated maximum polarization and dispersion due to its higher $\epsilon_1(\omega)$ value among other compounds. However, the following trend is CsZrH₃ > RbZrH₃ > KZrH₃ > NaZrH₃ > LiZrH₃, respectively. It is suggested that these materials have hard and bond-bending characteristics, leading to enhanced light absorption. Fig. 16 (d) shows the $\epsilon_2(\omega)$ component; high-intense peaks were notably observed in the energy range from 5 to 10 eV, respectively. These findings emphasize that all AeZrH₃ exhibited dielectric functions

relatively similar to each other, due to electronic polarizability and density of charge carriers. This characteristic holds that AeZrH₃ materials have significance across various applications, including optoelectronics and energy storage [71–74].

The optical $n(\omega)$ is a crucial parameter that dictates the transparency of a material and influences the bending or slowing down of light as it traverses through. This index is a composite of two components: the real part, denoted as $n(\omega)$, which signifies the materials transparency to incident radiation. The imaginary part, denoted as $k(\omega)$, provides insights into the materials propensity to absorb radiation at specific wavelengths [70]. Fig. 16(a) illustrates that the real part of the refractive index $n(\omega)$ value is 3.8 (LiZrH₃), 3.4 (NaZrH₃), 4.8 (RbZrH₃), whereas the KZrH₃ and CsZrH₃ are higher than 7, respectively, at zero photon energy. whereas the imaginary refractive index $k(\omega)$ maximum peak values were noticed at 5–10 eV energy range, respectively, presented in Fig. 16 (b). These results suggested that AeZrH₃ materials hold promise for enhancing applications in optoelectronics and photonics [75–79].

Furthermore, the optical loss function $L(\omega)$ factor pertains to the loss of radiation attributed to the scattering and dispersion of photons. The $L(\omega)$ is directly correlated with the likelihood of scattered radiation [80, 81]. Fig. 17 (a) shows that the maximum peak values are 15 to 20 eV range of energy for AeZrH₃ (Ae = Li/Na/K/Rb/Cs) compounds. In this case, the LiZrH₃ shows a highly intense loss function peak. Indicated that it is a huge loss as compared to the rest hydrides. Furthermore, these results suggested that AeZrH₃ is closely related to each compound due to slightly different electronic behavior and bonding interactions between atoms. Nevertheless, the sudden fall or rise is plasmonic peak [80]. The optical conductivity $\sigma(\omega)$ plays a pivotal role in materials intended for optoelectronic applications. It encompasses the conduction mechanism of photoelectrons within a material, describing incident photon interaction with the surface of a compound [81].

The absorption of electromagnetic radiation by the AeZrH₃ compounds is governed by the absorption coefficient, $\alpha(\omega)$ [82]. As expected, no absorption is observed at 0 eV. However, as the photon energy increases, the absorption also increases, as illustrated in Fig. 17

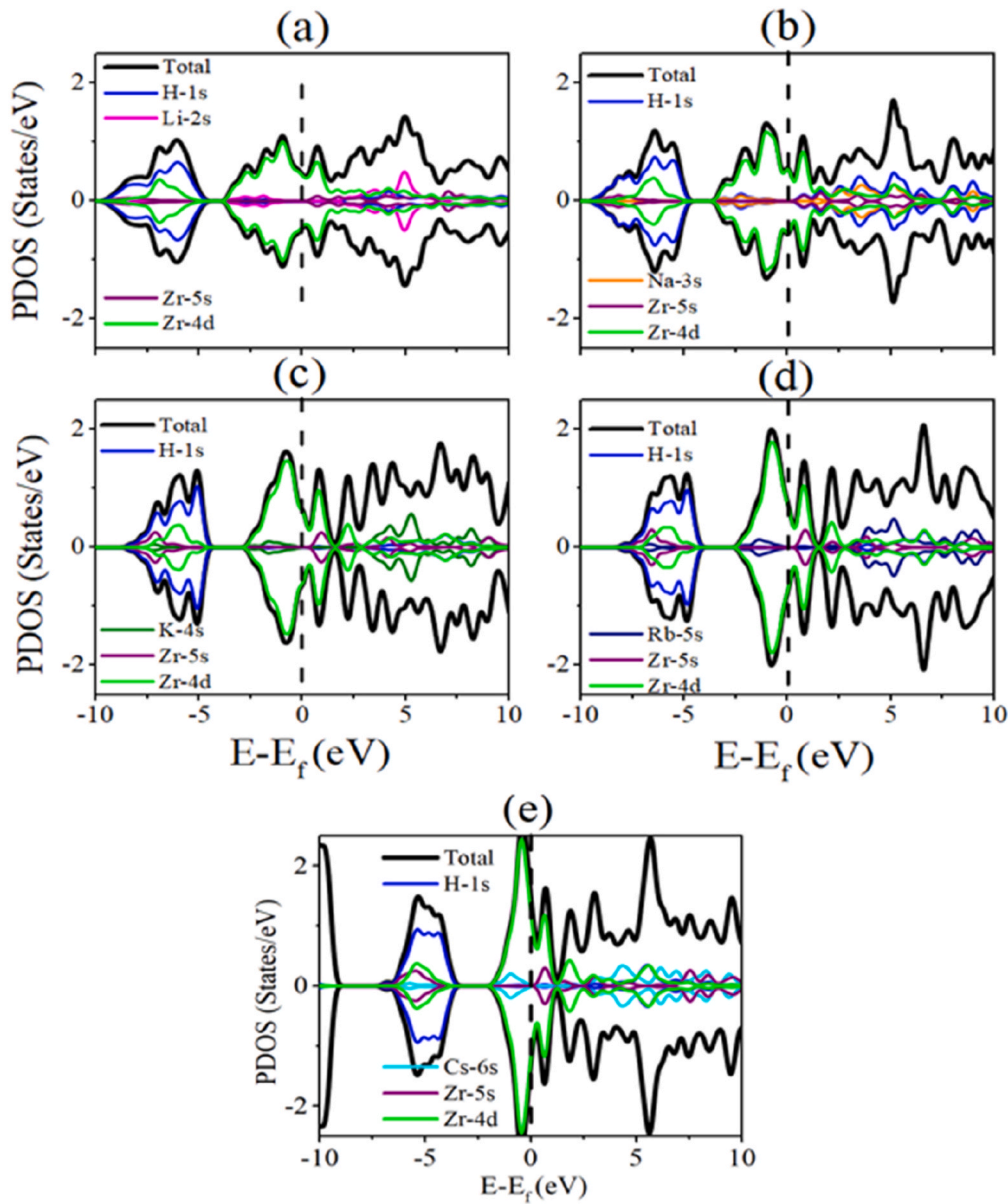


Fig. 13. The partial density of state of (a) LiZrH₃, (b) NaZrH₃, (c) KZrH₃, (d) RbZrH₃, and (e) CsZrH₃ hydrides, without SOC.

(b). The materials exhibit strong absorption peaks in the range of energy are 5 to 20 eV. These results validate that all these hybrids show strong absorption in UV-visible light energy. However, among these, LiZrH₃ hydride demonstrates the highest absorption, which can be attributed to enhanced electronic transitions that facilitate the movement of electrons between energy levels when photons are absorbed. Interestingly, the AeZrH₃ compounds exhibit the same trends the maximum absorption decreasing as the substitution of Li to Cs. Moreover, the optical absorption shows the metallic behavior, which agrees with the electronic structure characteristics (Fig. 10). Although A high absorption coefficient is particularly beneficial for applications such as energy harvesting, photodetectors, sensors, optoelectronic devices, and energy storage technologies [12,83–85].

The optical reflectivity, $R(\omega)$, represents the proportion of incident light reflected from the materials surface [81]. Fig. 17(c) displays the reflectance spectra of AeZrH₃ compounds. At zero photon energy ($\omega = 0$), the reflectivity values are 35%, 28%, 70%, 50%, and 63% for LiZrH₃, NaZrH₃, KZrH₃, RbZrH₃, and CsZrH₃, respectively. Maximum reflectance values are observed at in the range of energy at 5 to 20 eV, whereas the trend follows as LiZrH₃ > NaZrH₃ > KZrH₃ > RbZrH₃ > CsZrH₃,

respectively, due the Li to Cs is the high loss and absorption. However, the same behavior was noticed in the rest optical characteristics. Furthermore, after 20 eV, the peak reflectivity tends to decrease with increasing photon energy. Materials with high reflectivity are desirable in solar cell applications due to their ability to reflect and manage light effectively [86].

Fig. 17(d) illustrates the real part of the optical conductivity of AeZrH₃ compounds. The data indicate that optical conductivity generally decreases with increasing photon energy, which is mainly due to plasmon emission effects. Notably, LiZrH₃ shows superior optical conductivity compared to the other compounds and follows the same behavior in the rest of the optical properties. This behavior is linked to strong absorption peaks, availability of unoccupied electronic states, and differences in the density of states [80]. High optical conductivity enhances the performance of materials in optoelectronic and energy-related applications [12,80,87].

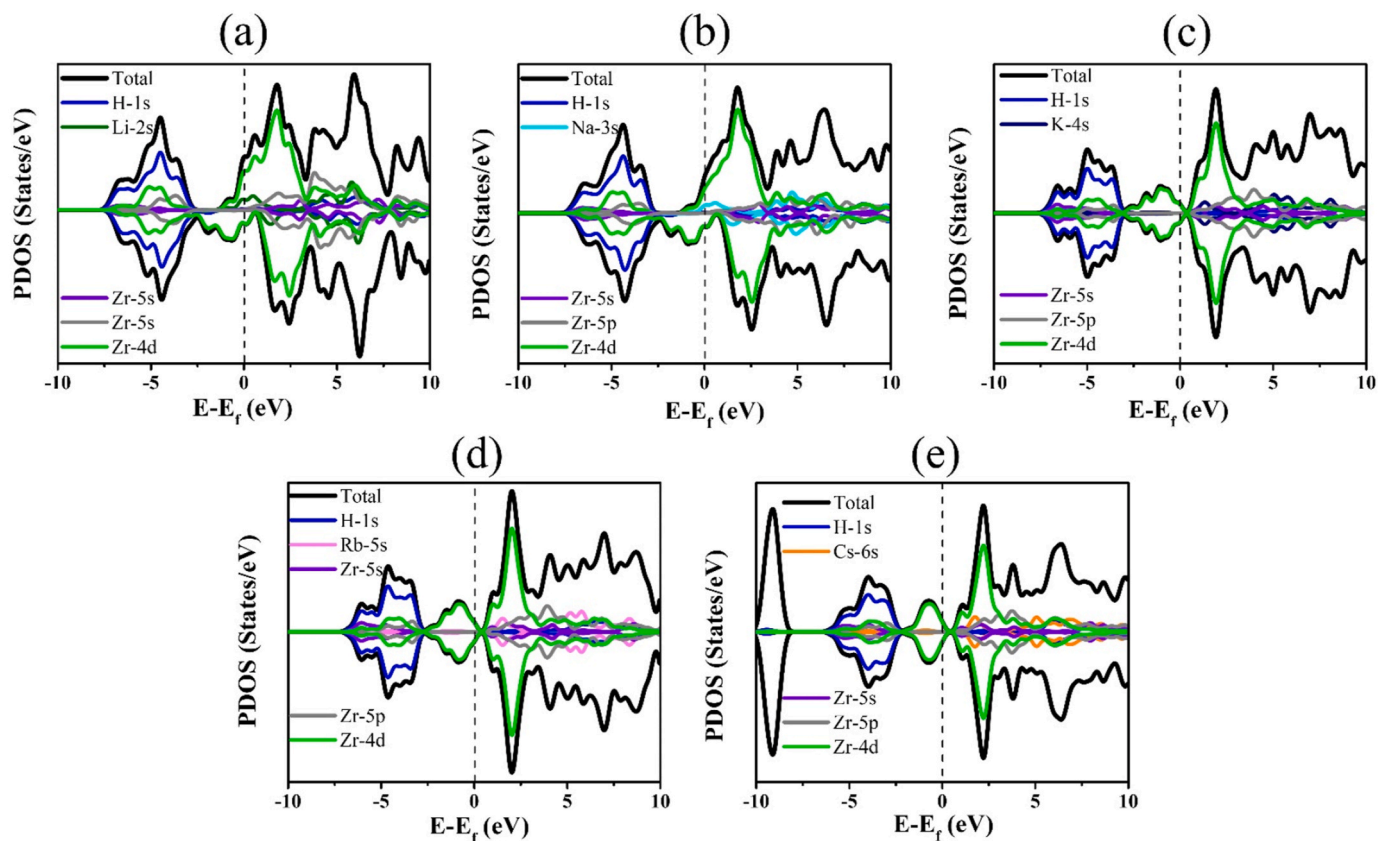


Fig. 14. The plotted graphs of spin-polarized partial density of states (PDOS) of (a) LiZrH₃, (b) NaZrH₃, (c) KZrH₃, (d) RbZrH₃ and (e) CsZrH₃ light materials with SOC and DFT + U method.

Table 4

The calculated atomic populations and Mulliken charges of AeZrH₃ materials.

Material	Species	Ion	s	p	d	Total	Charge e
LiZrH ₃	H	1	1.33	0.00	0.00	1.33	-0.33
	H	2	1.33	0.00	0.00	1.33	-0.33
	H	3	1.33	0.00	0.00	1.33	-0.33
	Li	1	2.15	0.00	0.00	2.15	0.85
	Zr	1	2.27	6.63	2.97	11.87	0.13
NaZrH ₃	H	1	1.36	0.00	0.00	1.36	-0.36
	H	2	1.36	0.00	0.00	1.36	-0.36
	H	3	1.36	0.00	0.00	1.36	-0.36
	Na	1	2.23	5.89	0.00	8.12	0.88
	Zr	1	2.37	6.16	2.80	11.79	0.21
KZrH ₃	H	1	1.48	0.00	0.00	1.48	-0.48
	H	2	1.48	0.00	0.00	1.48	-0.48
	H	3	1.48	0.00	0.00	1.48	-0.48
	K	1	1.96	6.02	0.00	7.98	1.02
	Zr	1	2.61	6.16	2.80	11.58	0.42
RbZrH ₃	H	1	1.46	0.00	0.00	1.46	-0.46
	H	2	1.46	0.00	0.00	1.46	-0.46
	H	3	1.46	0.00	0.00	1.46	-0.46
	Rb	1	1.96	5.79	0.00	7.75	1.25
	Zr	1	2.71	6.36	2.80	11.86	0.14
CsZrH ₃	H	1	1.47	0.00	0.00	1.47	-0.47
	H	2	1.47	0.00	0.00	1.47	-0.47
	H	3	1.47	0.00	0.00	1.47	-0.47
	Cs	1	1.56	6.30	0.00	7.86	1.14
	Zr	1	2.71	6.13	2.88	11.72	0.28

4. The role of the d band-center and hydrogen storage application

According to the d-band theory [88], the electronic density of states (DOS) near the Fermi level plays a pivotal role in governing the

adsorption energy between a metal surface and an adsorbate. This interaction primarily stems from the hybridization between the d-orbitals of the metal atoms and the orbitals of the adsorbate. As a result, any alteration in the d-band electronic structure (d-DOS) can substantially impact both the strength of adsorption and the overall catalytic activity of the surface. To characterize these electronic features, the d-band center model is widely adopted. This model correlates the position of the d-band center relative to the Fermi level with the adsorption behavior of the material. One critical factor that influences the position of the d-band center is lattice strain. Strain modifies the interatomic distances within the metal lattice, thereby affecting the overlap of d-orbitals. These changes can lead to either the narrowing or broadening of the d-band, and subsequently shift the d-band center closer to or further from the Fermi level. Numerous studies using DFT have established clear relationships between strain-induced modifications in the electronic structure and variations in catalytic efficiency, lending strong support to the validity of the d-band center model [88,89].

In our study, we calculated the d-band centers for the AeZrH₃ (Ae = Li, Na, K, Rb, Cs) compounds, as illustrated in Fig. 18. The results show that substituting Li with heavier alkali metals such as Na, K, Rb, and Cs leads to a reduction in hydrogen storage capacity. Interestingly, this substitution results in a rise in the d-band center position. Notably, in these hydrides, the Zr d-band features unoccupied states, making them capable of adsorbing hydrogen atoms. A shift of the d-band center toward the Fermi level is associated with enhanced catalytic activity and stronger interaction with hydrogen. Therefore, our findings support the d-band theory and highlight the promising potential of these materials for hydrogen storage applications.

Among various candidates for hydrogen storage, metal hydrides have emerged as some of the most promising materials due to their high storage capacities and favorable thermodynamics [5,90]. Magnesium hydride (MgH₂), in particular, has attracted widespread attention owing

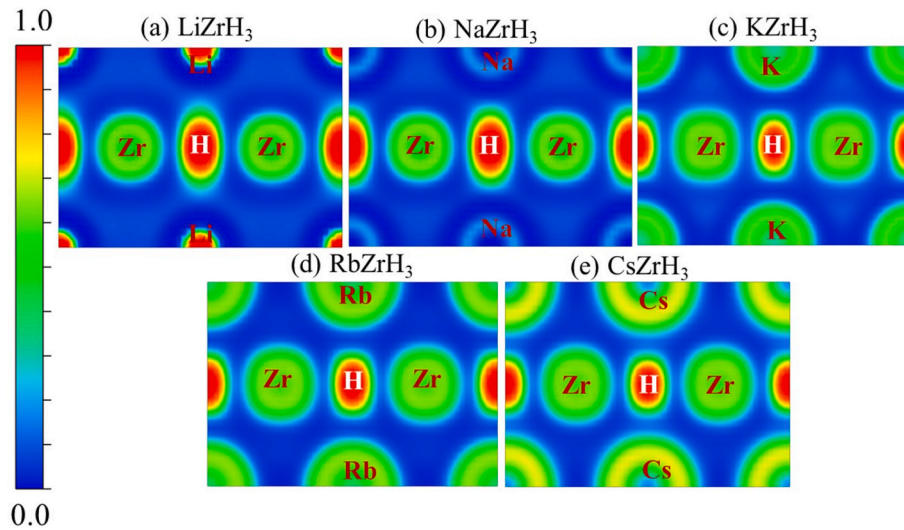


Fig. 15. Electron localization function of $AeZrH_3$ materials along with (110) plane orientation.

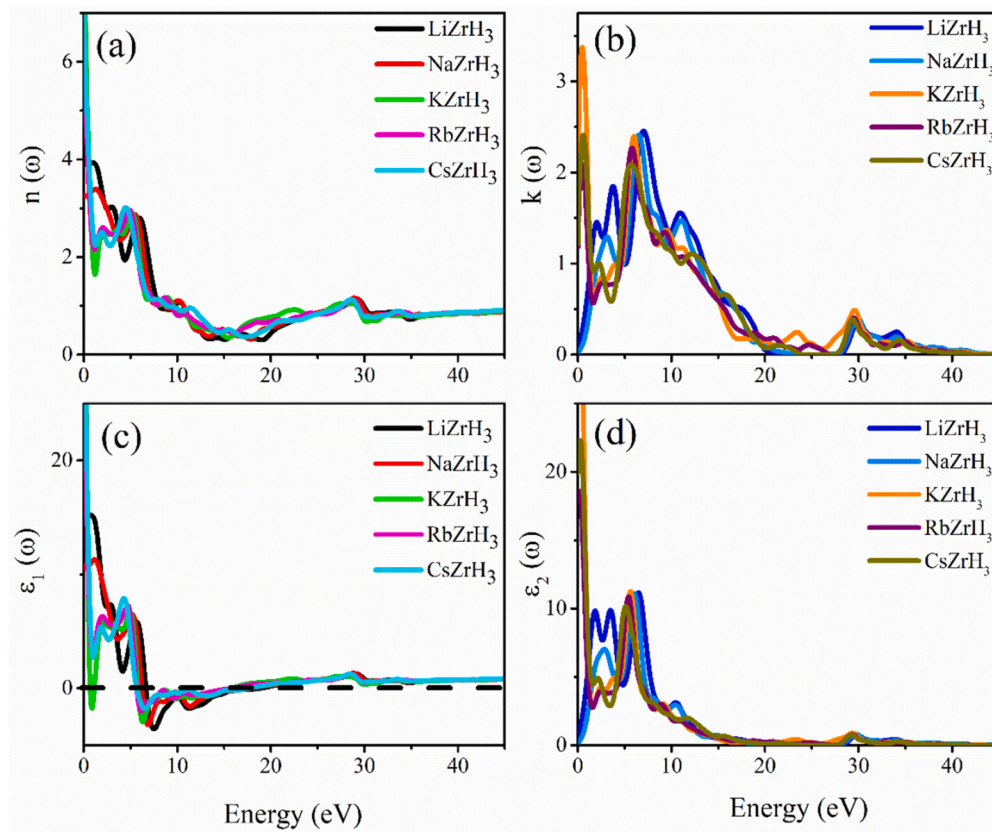


Fig. 16. The plotted graphs of (a) refractive index of real components, (b) refractive index of imaginary components, (c, d) dielectric function of real and imaginary components of $AeZrH_3$ materials.

to its high reversible hydrogen storage capacity (7.6 wt%), low cost, and lightweight characteristics [91]. Despite these advantages, its practical application is hindered by slow hydrogen sorption kinetics and a relatively high formation enthalpy [71]. To overcome these limitations, recent studies have shifted focus towards ternary hydrides, specifically those with the general formula ABH_3 . These materials belong to the broader class of perovskite compounds, typically represented as ABX_3 . Ternary hydrides have shown promising hydrogen storage properties, offering moderate to high gravimetric capacities (ranging from 1.2 to

6.0 wt%), along with good thermal stability and tunable kinetics [92]. The perovskite hydride structure usually features an alkali or alkaline earth metal at the A-site, a transition or post-transition metal at the B-site, and hydrogen atoms occupying the X-site [48]. This unique framework supports reversible hydrogen absorption and desorption via solid-state reactions. Furthermore, the perovskite structure allows for two oxidation state configurations $A^+B^{2+}H_3^-$ and $A^{2+}B^+H_3^-$, which facilitate efficient hydrogen incorporation within the lattice. Specifically, the hydrogen gravimetric capacities of $AeZrH_3$ perovskite-type

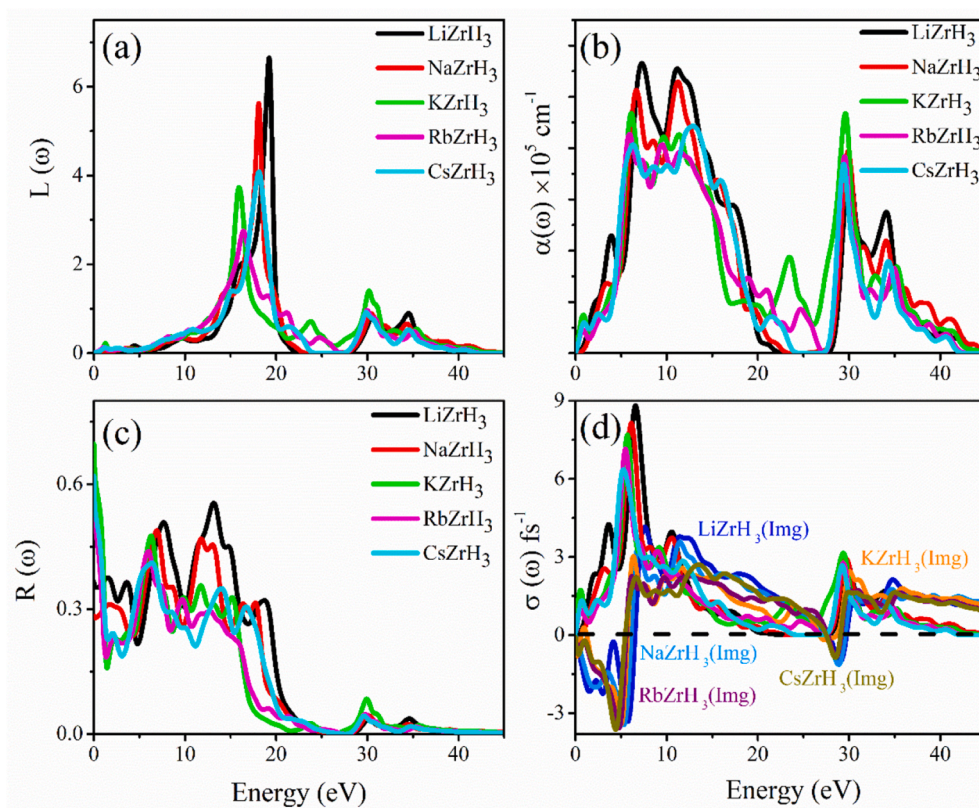


Fig. 17. The plotted graphs of (a) loss function, (b) absorption coefficient, (c) reflectivity, and (d) conductivity components of AeZrH₃ materials.

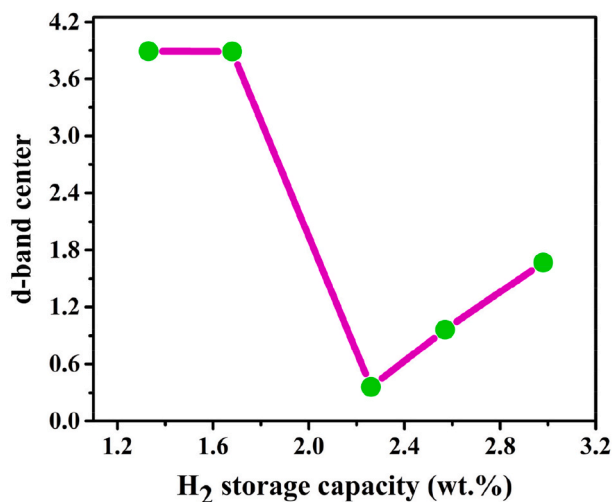


Fig. 18. The relation between the d-band center and hydrogen storage capacity for AeZrH₃ materials.

hydrides have been evaluated using a defined equation (eq-28) [14], offering insights into their potential for practical hydrogen storage applications.

$$C_{wt.\%} = \left(\frac{\frac{H}{M} M_H}{\frac{H}{M} M_H + M_{Host}} \times 100 \right) \% \quad (28)$$

In this study, the ratio of hydrogen atoms to host material atoms (H/M), the molar mass of the hydride (M_H), and the molar mass of the host material (M_{Host}) were used to calculate the gravimetric hydrogen storage capacity. The calculated hydrogen storage capacities for LiZrH₃,

NaZrH₃, KZrH₃, RbZrH₃, and CsZrH₃ were found to be 2.98, 2.57, 2.26, 1.68, and 1.33 wt%, respectively, as illustrated in Fig. 19. These results indicate a clear trend: the hydrogen storage capacity decreases as the atomic number of the alkali metal increases. This decline is attributed to the larger atomic mass and size of heavier alkali metals, which leads to stronger interatomic interactions, both factors that reduce hydrogen uptake efficiency per unit mass [48].

Remarkably, our findings outperform several previously reported ternary hydrides. For example, the storage capacities of KFeH₃ (2.99 wt%) [93], KPdH₃ (2.035 wt%) [67], Ba₂VH₆ (1.82 wt%) [94], RbFeH₃ (2.09 wt%) [95], and RbSrH₃ (1.33 wt%) [96] are all lower than those observed for LiZrH₃ and NaZrH₃ in our study. These promising results suggest that the investigated cubic perovskite hydrides, particularly those with lighter alkali elements, could serve as strong candidates for experimental development in hydrogen storage applications.

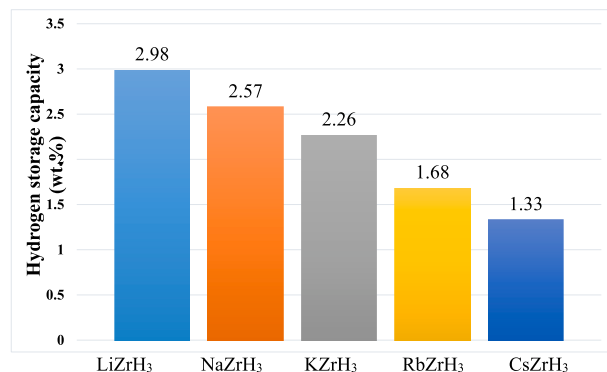


Fig. 19. The hydrogen storage capacity of AeZrH₃ materials.

5. Conclusion

In this study, we conducted a comprehensive investigation of novel perovskite-type hydrides, $AeZrH_3$ ($Ae = Li, Na, K, Rb, Cs$), as potential candidates for hydrogen storage, incorporating Van der Waals interactions within first-principles calculations. Our results reveal that all $AeZrH_3$ compounds exhibit metallic behavior, with and without SOC calculations. In contrast, under DFT + U calculations they exhibit semiconductor behavior, which is advantageous for ensuring high safety in hydrogen storage applications. Spin-polarized calculations confirmed that these materials are non-magnetic ($AeZrH_3$). Mechanical property analysis indicates that all the studied compounds are mechanically stable. Moreover, $LiZrH_3$, $RbZrH_3$, and $CsZrH_3$ exhibit ductile characteristics, whereas $NaZrH_3$ and $KZrH_3$ are classified as brittle. Thermodynamic evaluations provided estimates of Debye temperatures, melting points, and average sound velocities, supporting the mechanical findings. The hydrogen storage potential, measured as gravimetric capacity, was found to be 2.98, 2.57, 2.26, 1.68, and 1.33 wt% for $AeZrH_3$, respectively. Among these, $LiZrH_3$ stands out with the highest hydrogen storage capacity, along with excellent stability and safety features, highlighting its promise as a next-generation hydrogen storage material based on the perovskite structure.

CRediT authorship contribution statement

M. Kashif Masood: Investigation, Conceptualization. **Rania Charif:** Validation, Resources, Data curation. **Redi Kristian Pingak:** Visualization, Software, Data curation. **Asif Jamil:** Visualization, Validation, Investigation. **M. Sharif Kiyani:** Validation, Methodology, Investigation. **Tahani A. Alrebd:** Resources, Investigation, Formal analysis. **Javed Rehman:** Supervision, Methodology, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.calphad.2026.102944>.

Data availability

No data was used for the research described in the article.

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